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(54) **SUPPORT FRAME OF AN ADJUSTABLE
TARGET DISC-ENTRAPMENT DEVICE**

(52) **U.S. Cl.**

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(2013.01)

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(51) **Int. Cl.**

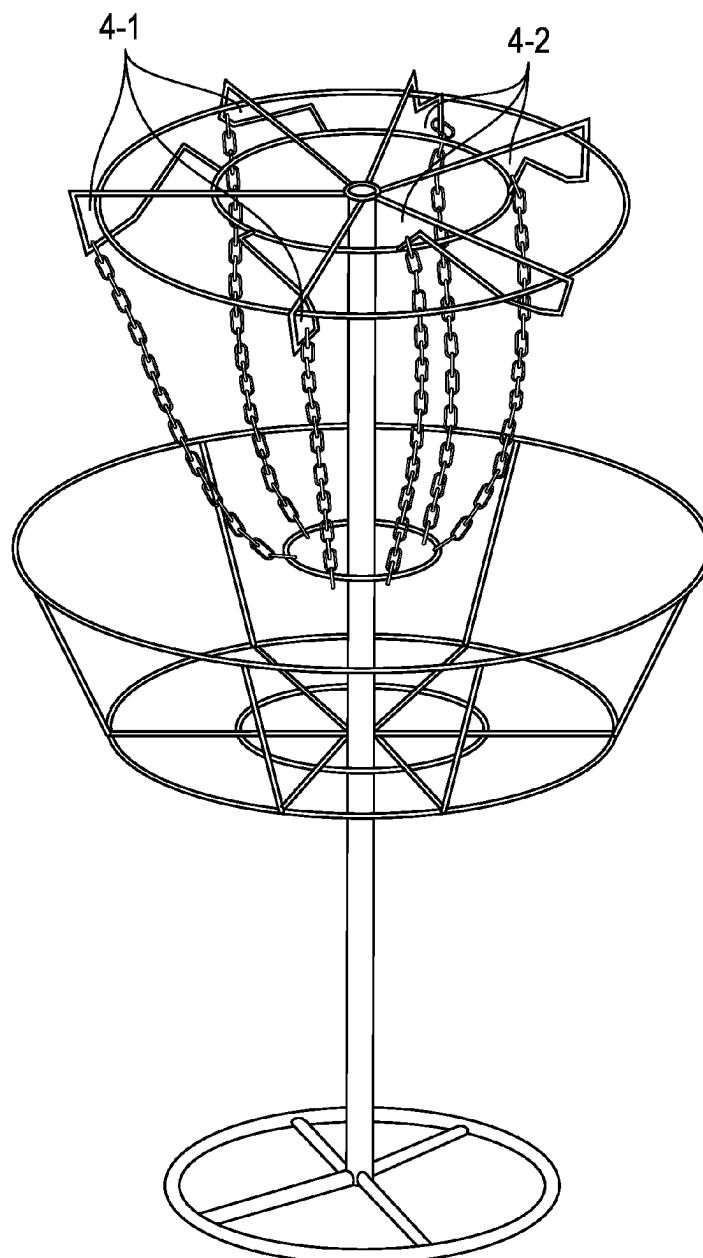
A63B 67/06

(2006.01)

(57)

ABSTRACT

An adjustable target disc-catching device allowing a player to manually slide a plurality of chains from outer chain wells to inner chain wells (or vice versa) allowing the player to easily contract or expand the target of the disc-catching device, thereby making a throw-in harder (contracting the chains to a smaller target area) or easier (expanding the chains to a larger target area).



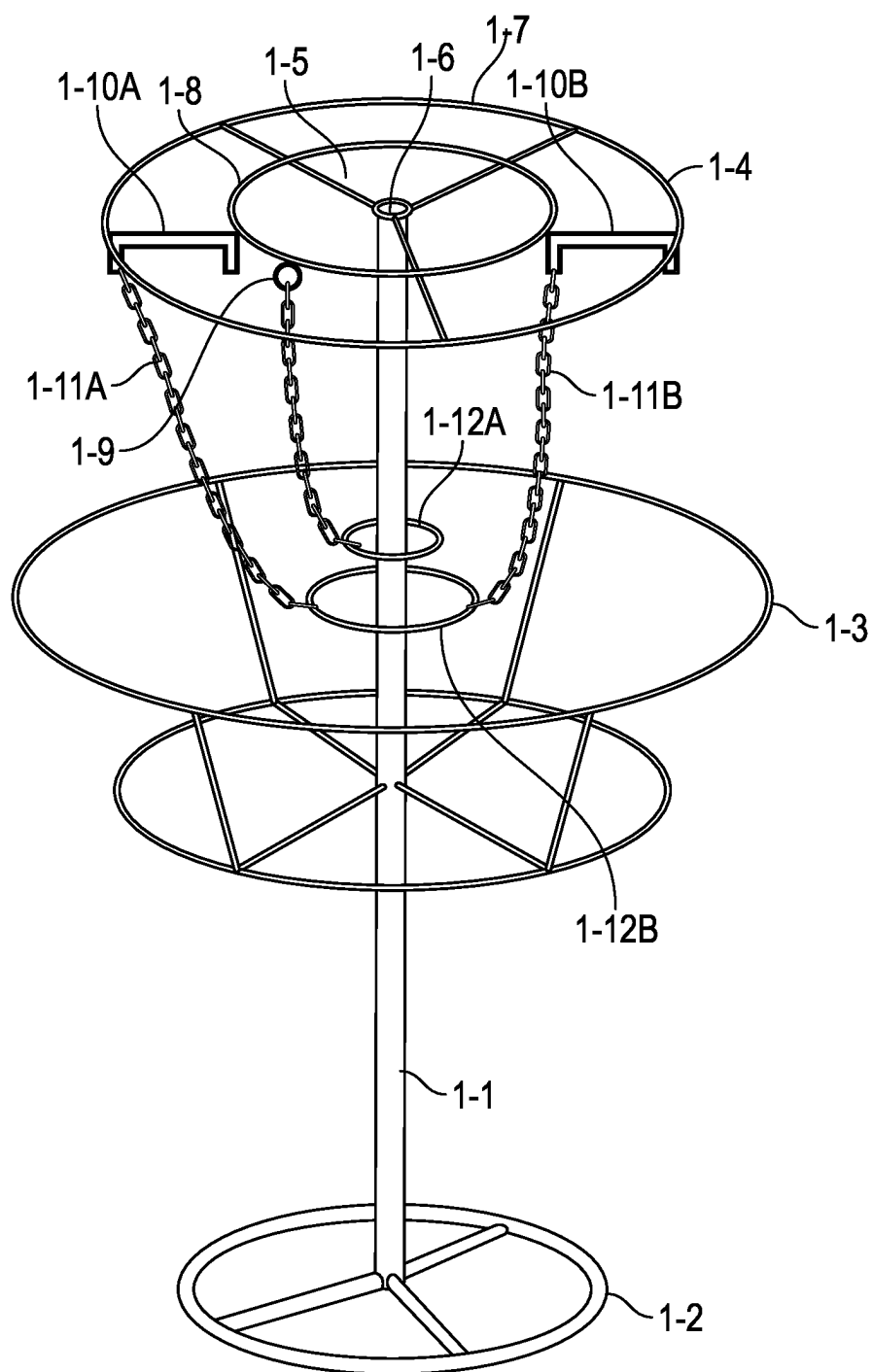


FIG. 1

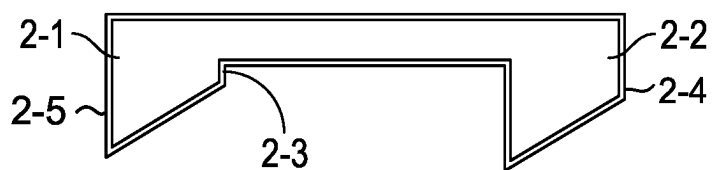


FIG. 2A

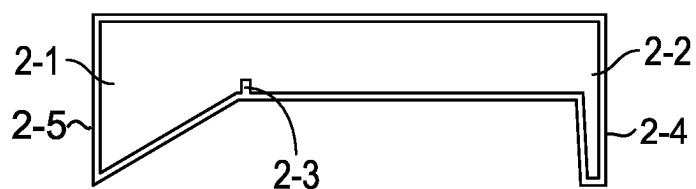


FIG. 2B

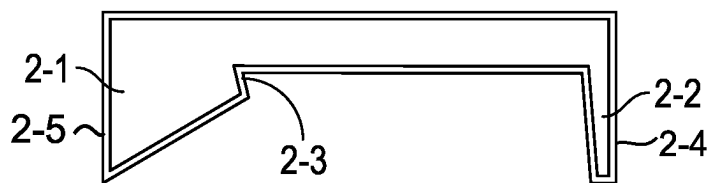


FIG. 2C

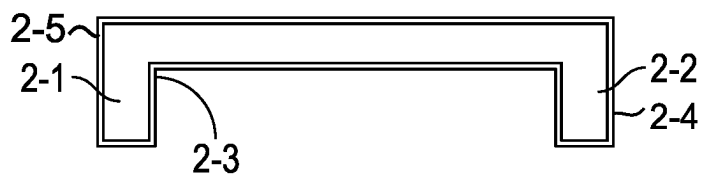


FIG. 2D

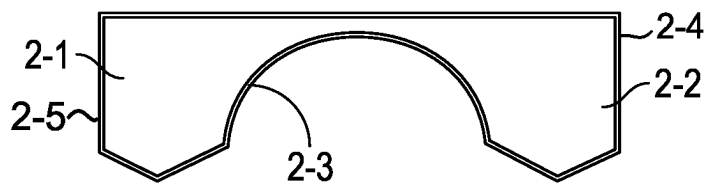


FIG. 2E

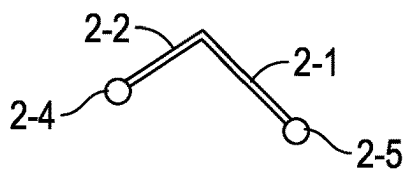


FIG. 2F

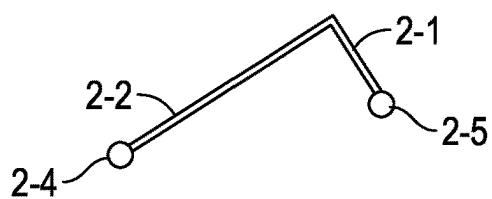


FIG. 2G

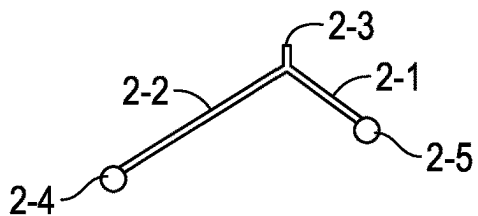


FIG. 2H

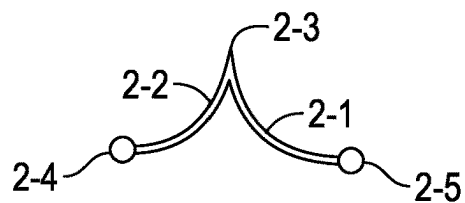


FIG. 2I

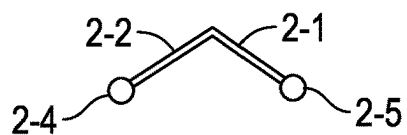


FIG. 2J

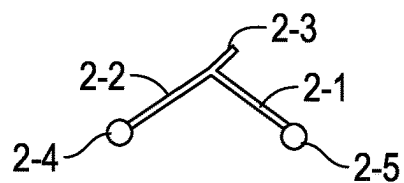


FIG. 2K

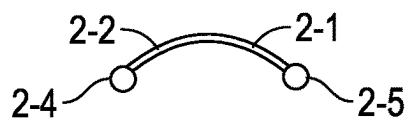


FIG. 2L

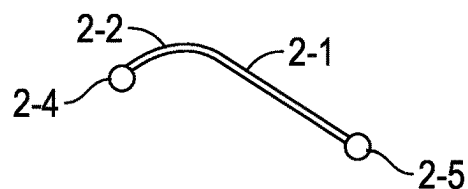


FIG. 2M

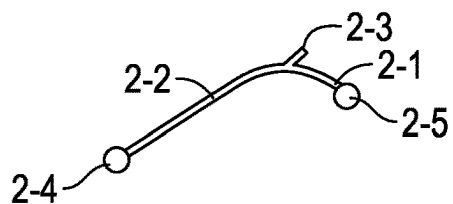


FIG. 2N

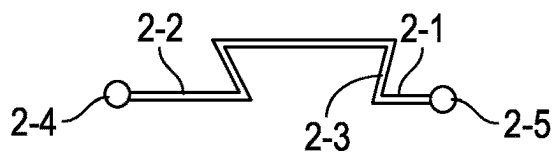


FIG. 2O

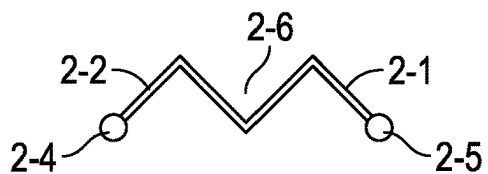


FIG. 2P

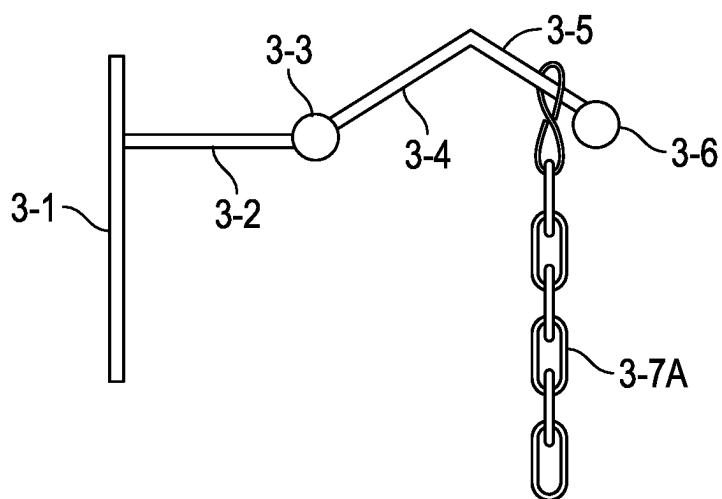


FIG. 3A

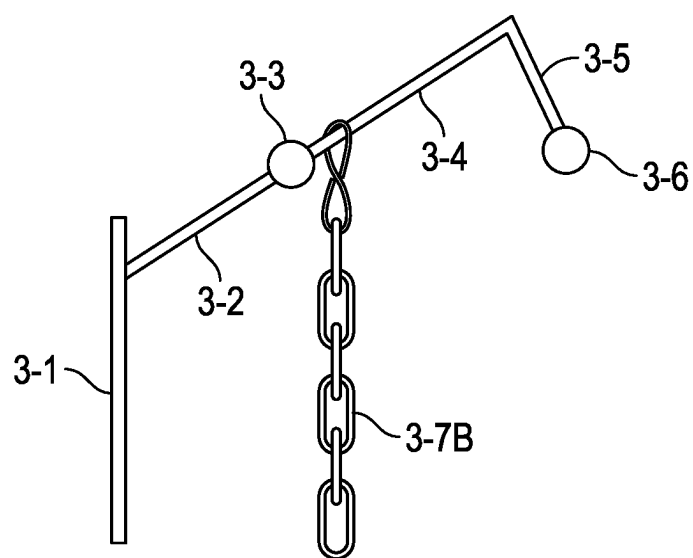


FIG. 3B

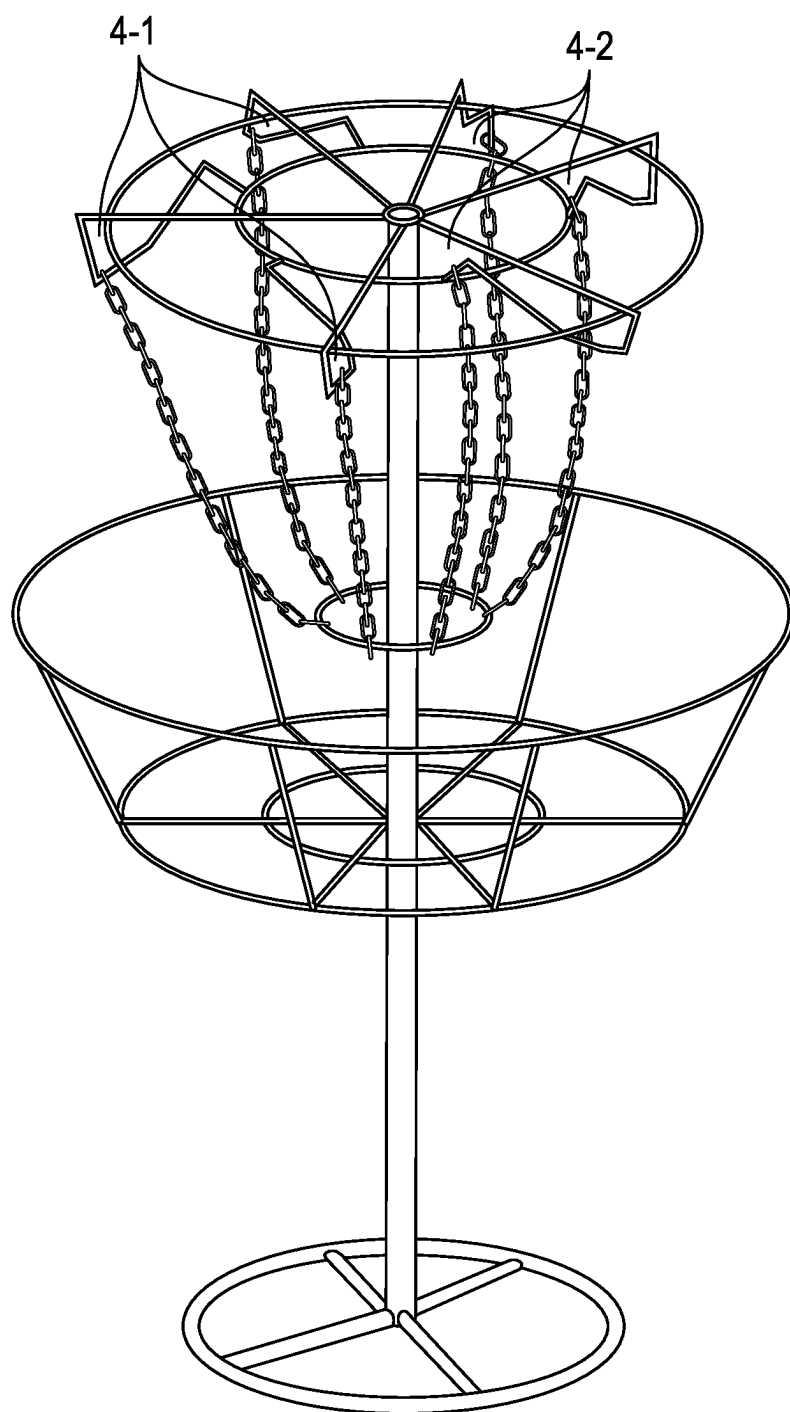


FIG. 4

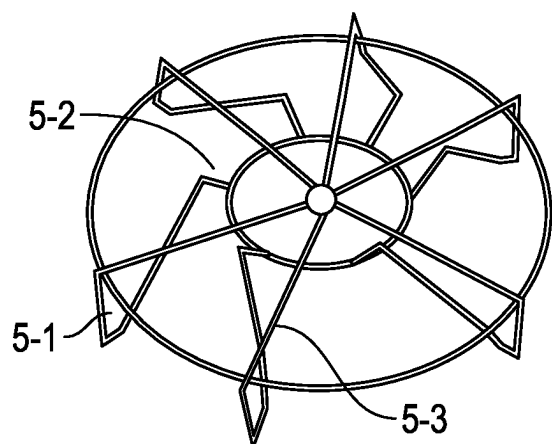


FIG. 5A

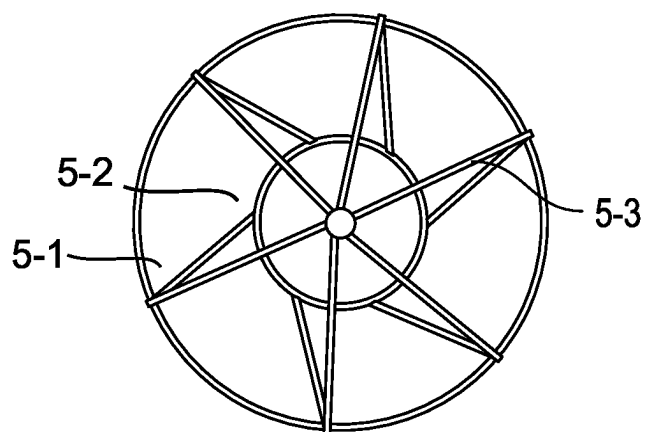


FIG. 5B

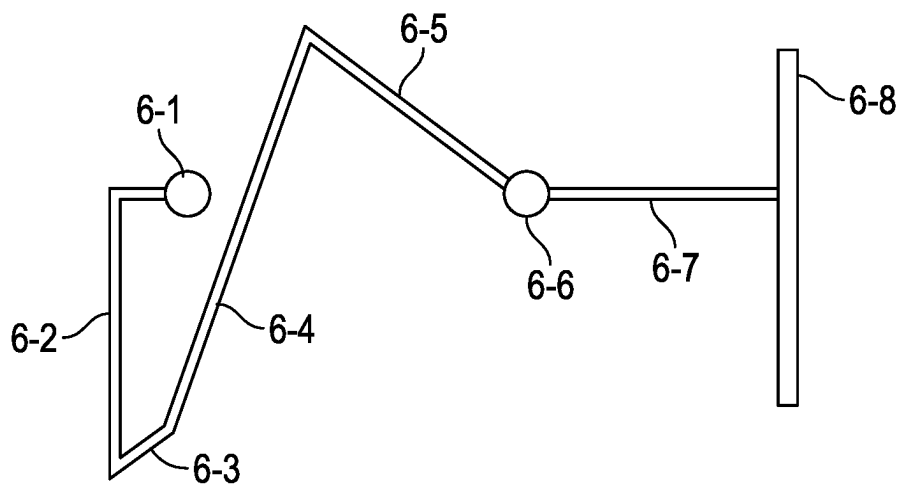


FIG. 6

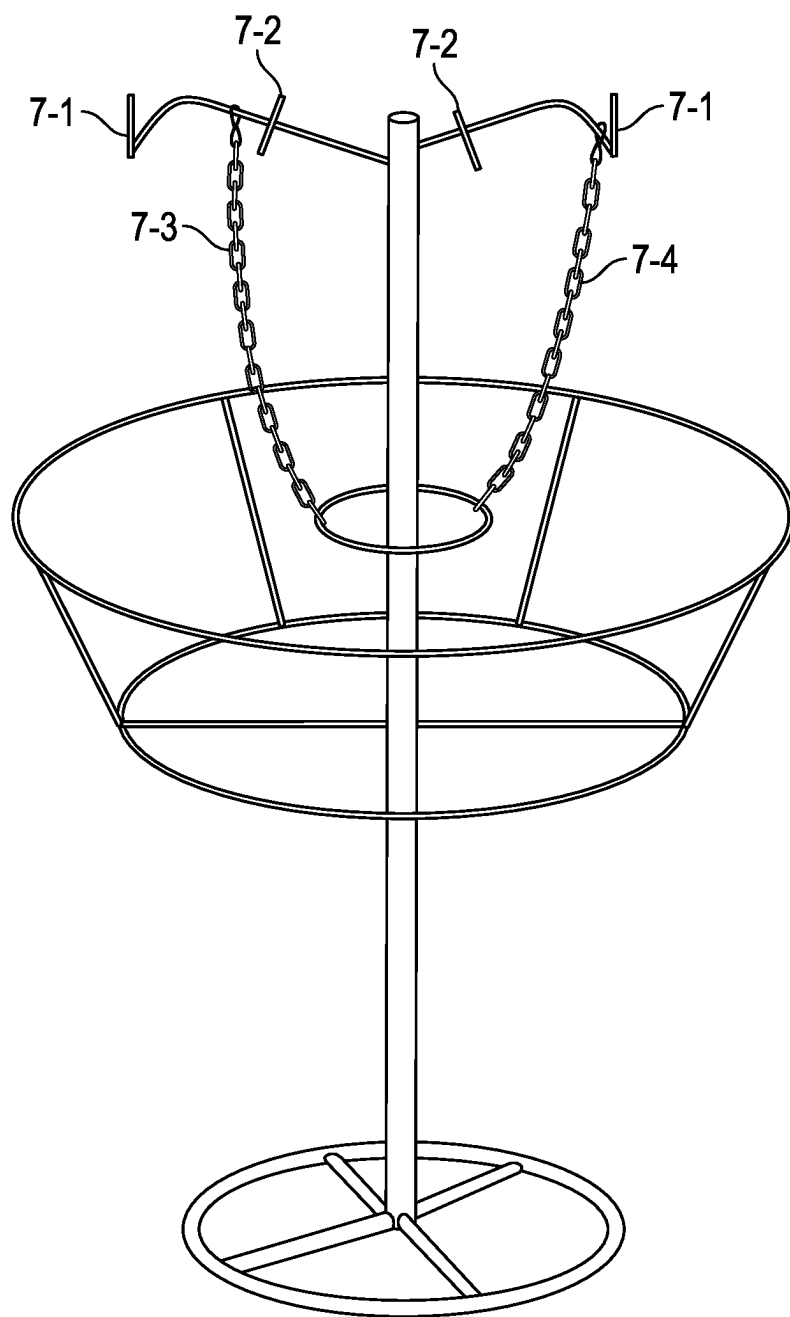


FIG. 7

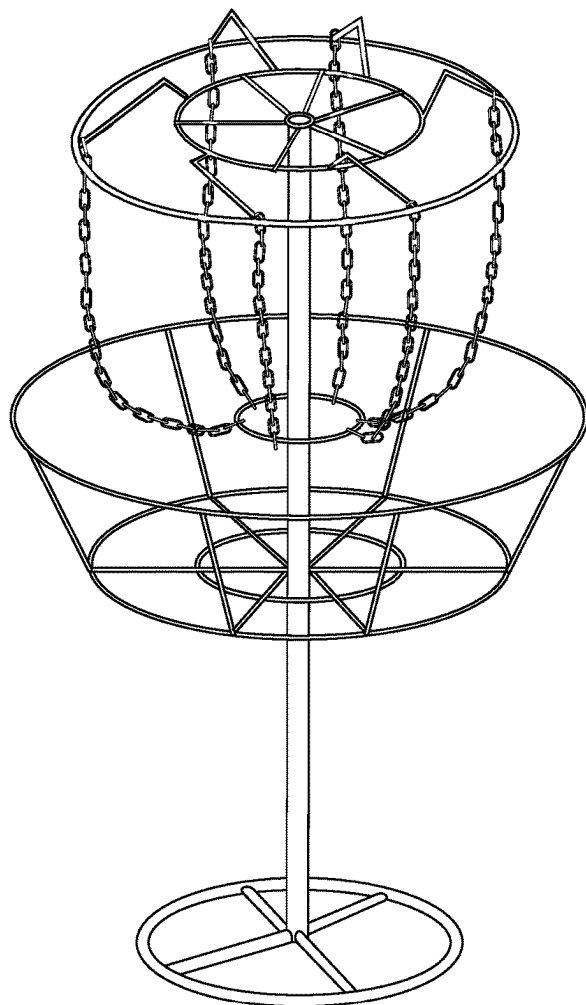


FIG. 8A

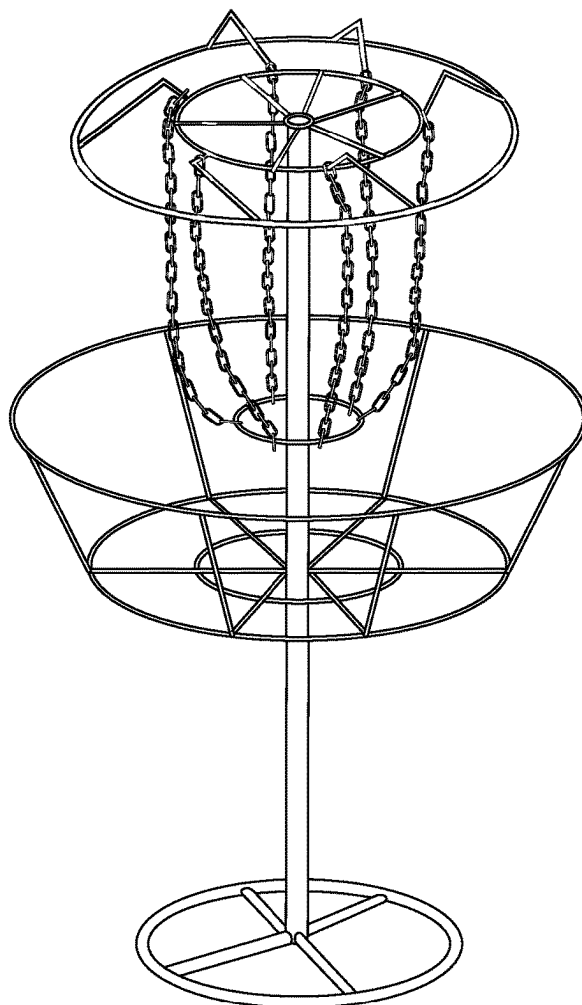


FIG. 8B

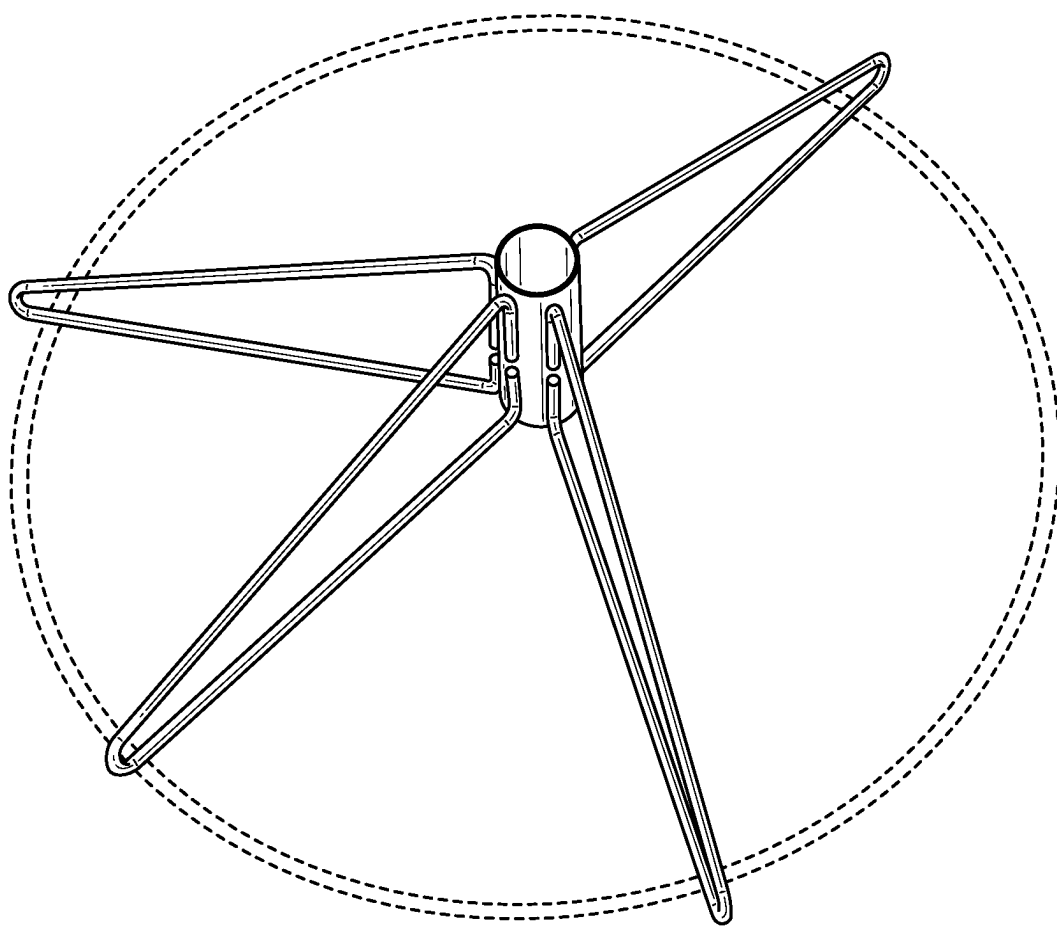


FIG. 9

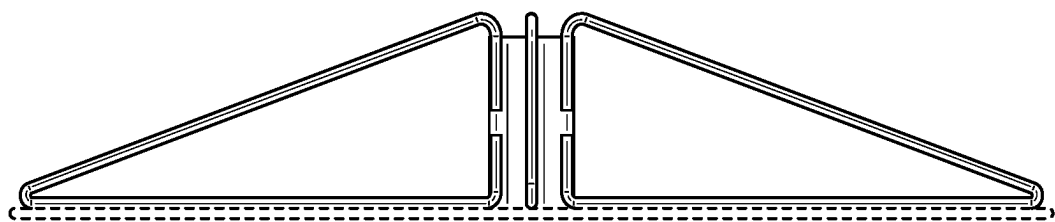


FIG. 10

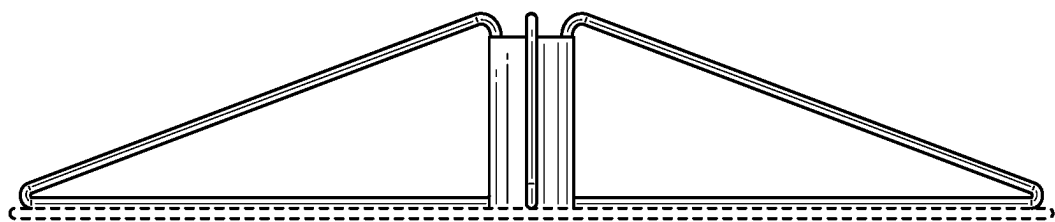


FIG. 11

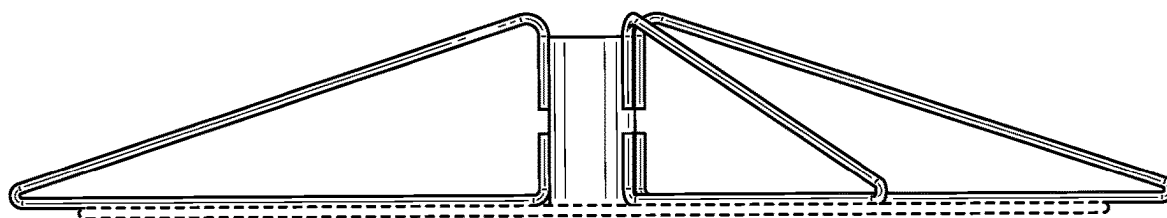


FIG. 12

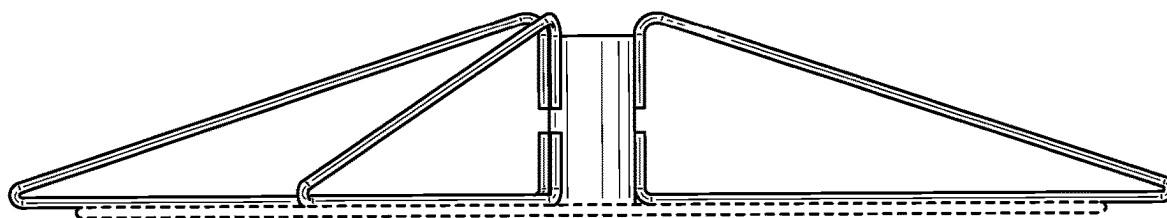


FIG. 13

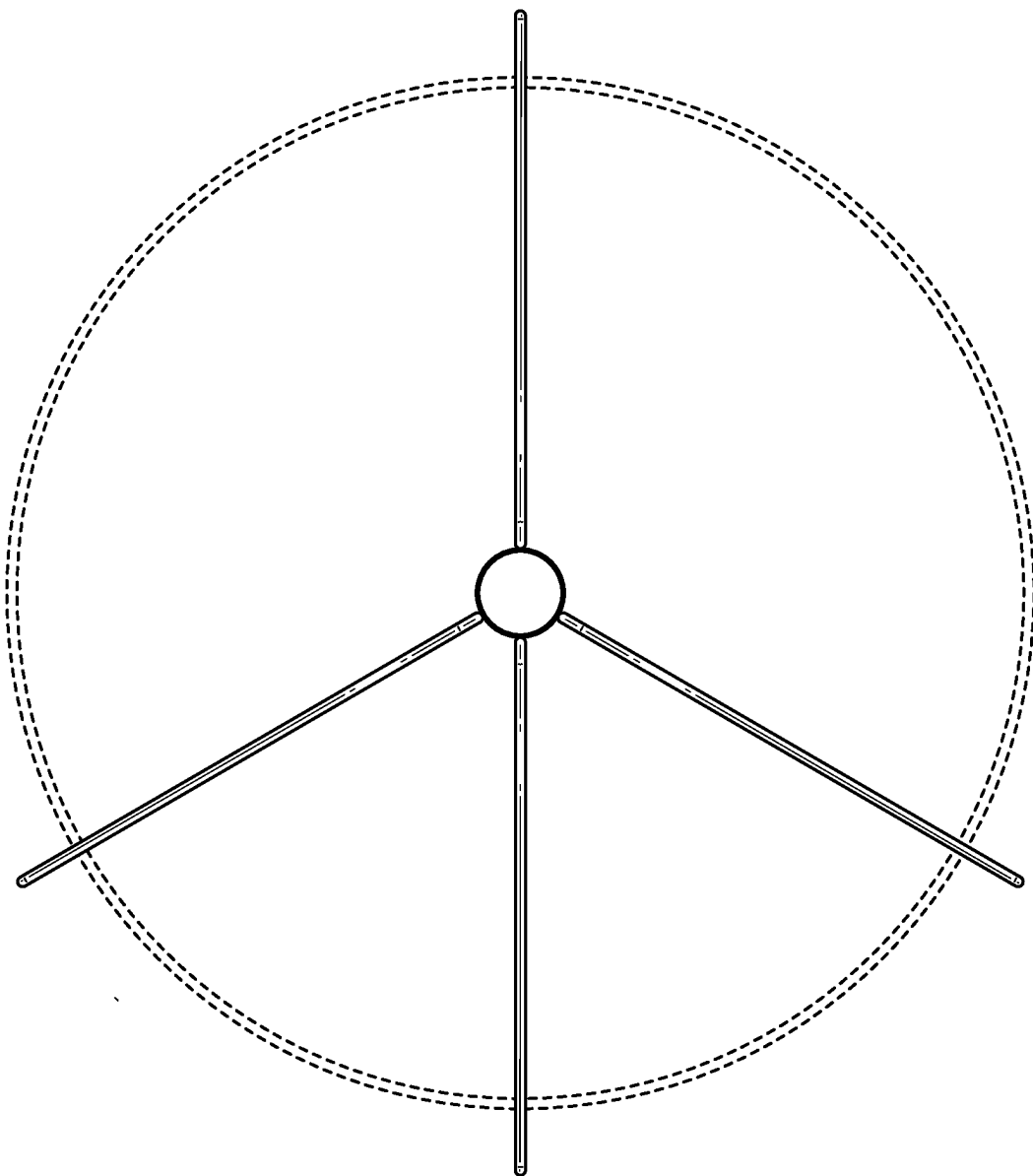


FIG. 14

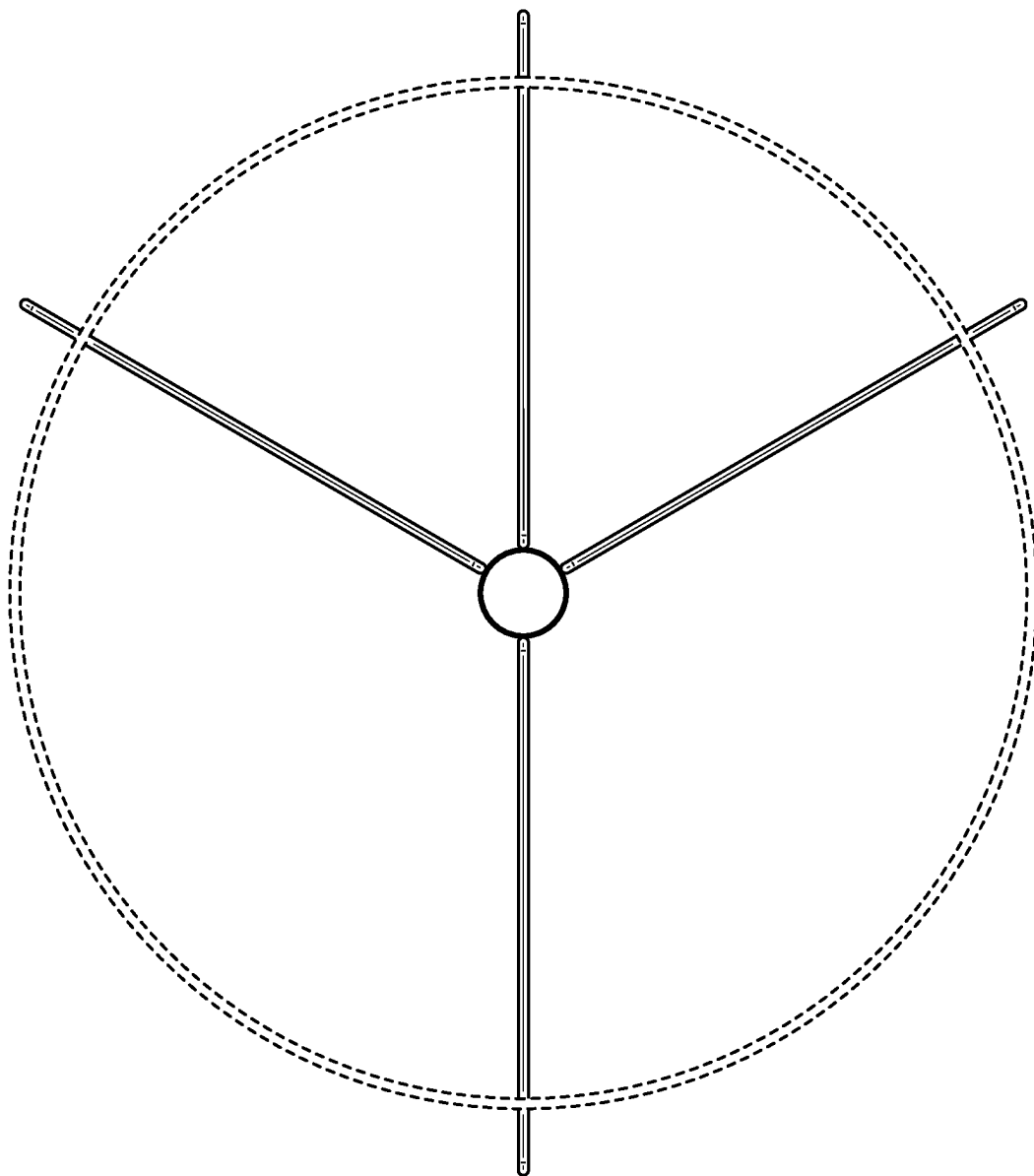


FIG. 15

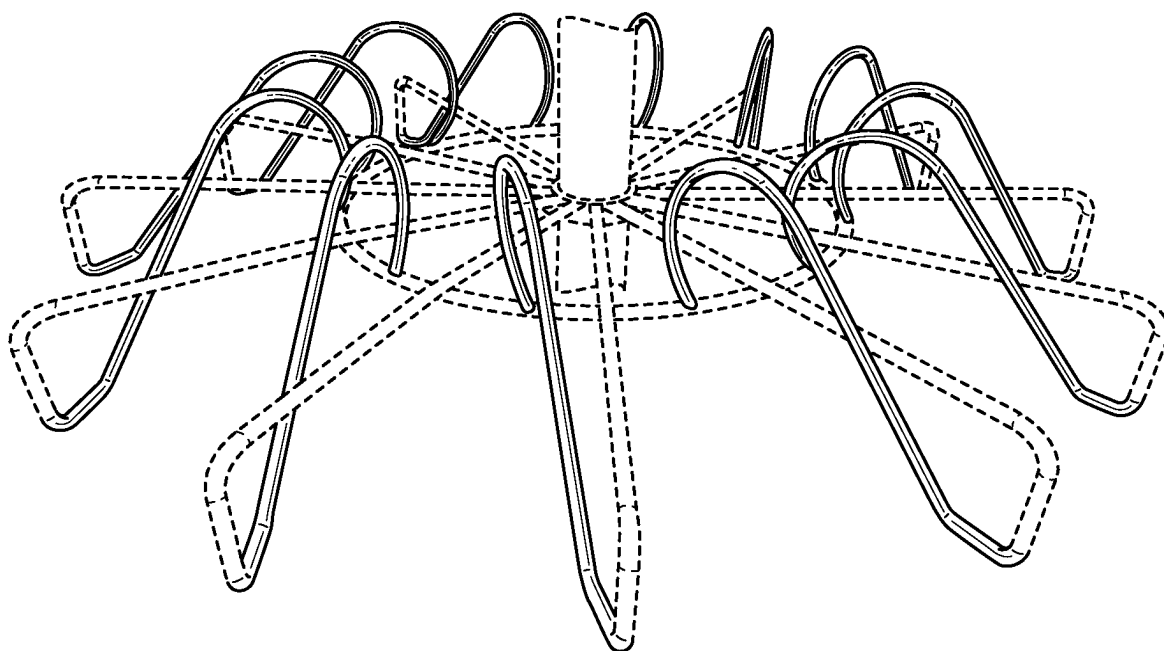


FIG. 16

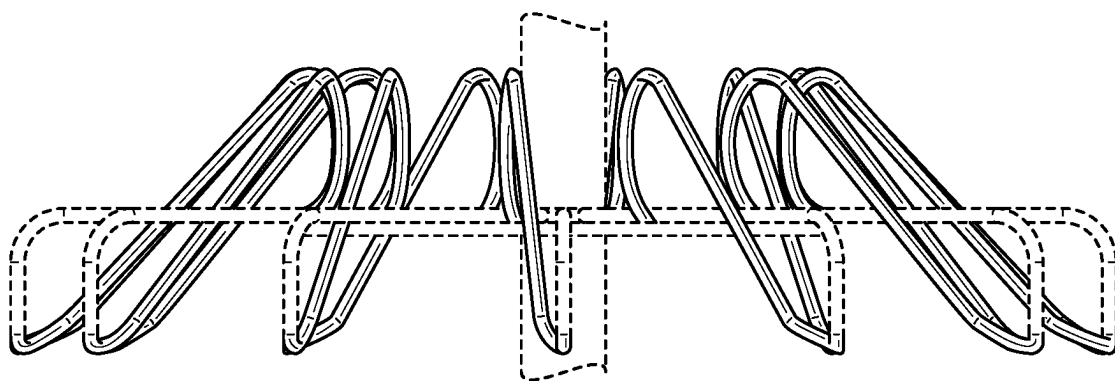


FIG. 17

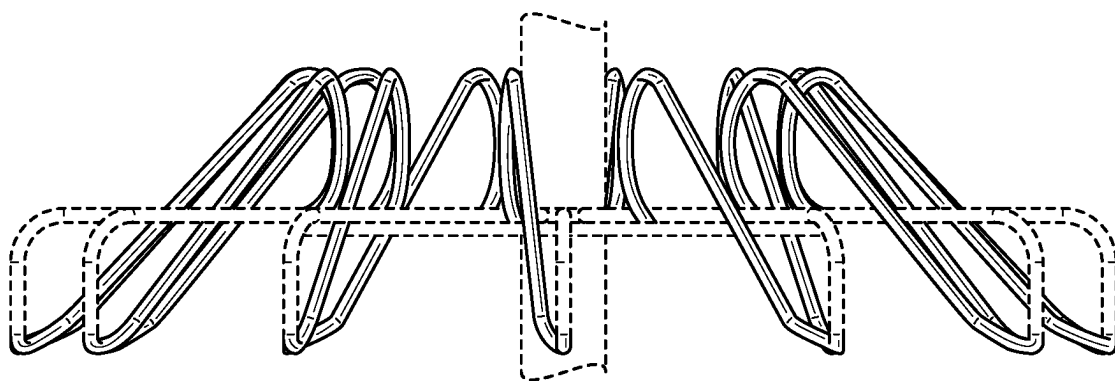


FIG. 18

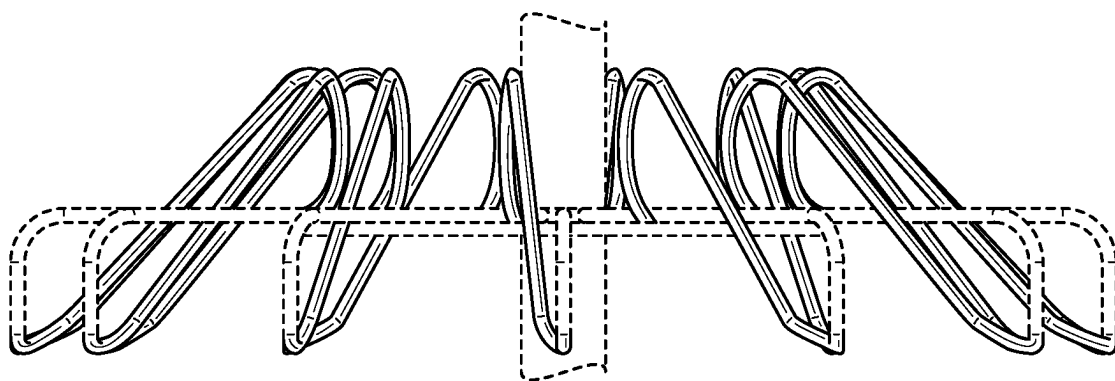


FIG. 19

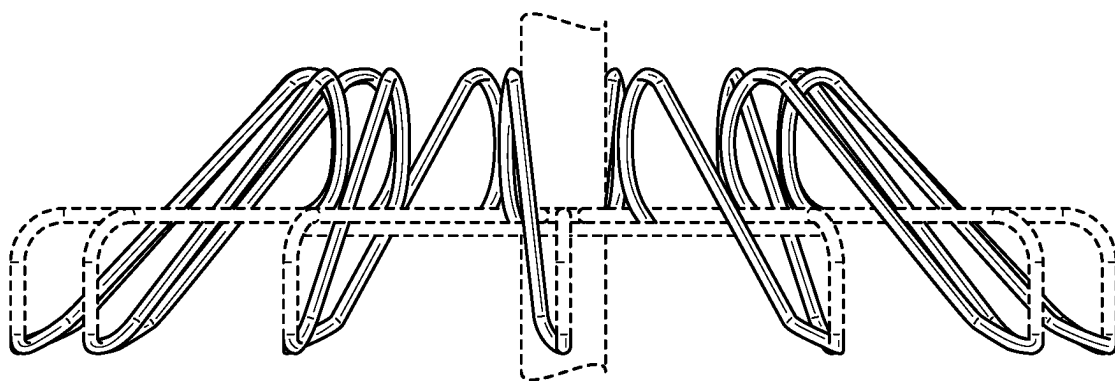


FIG. 20

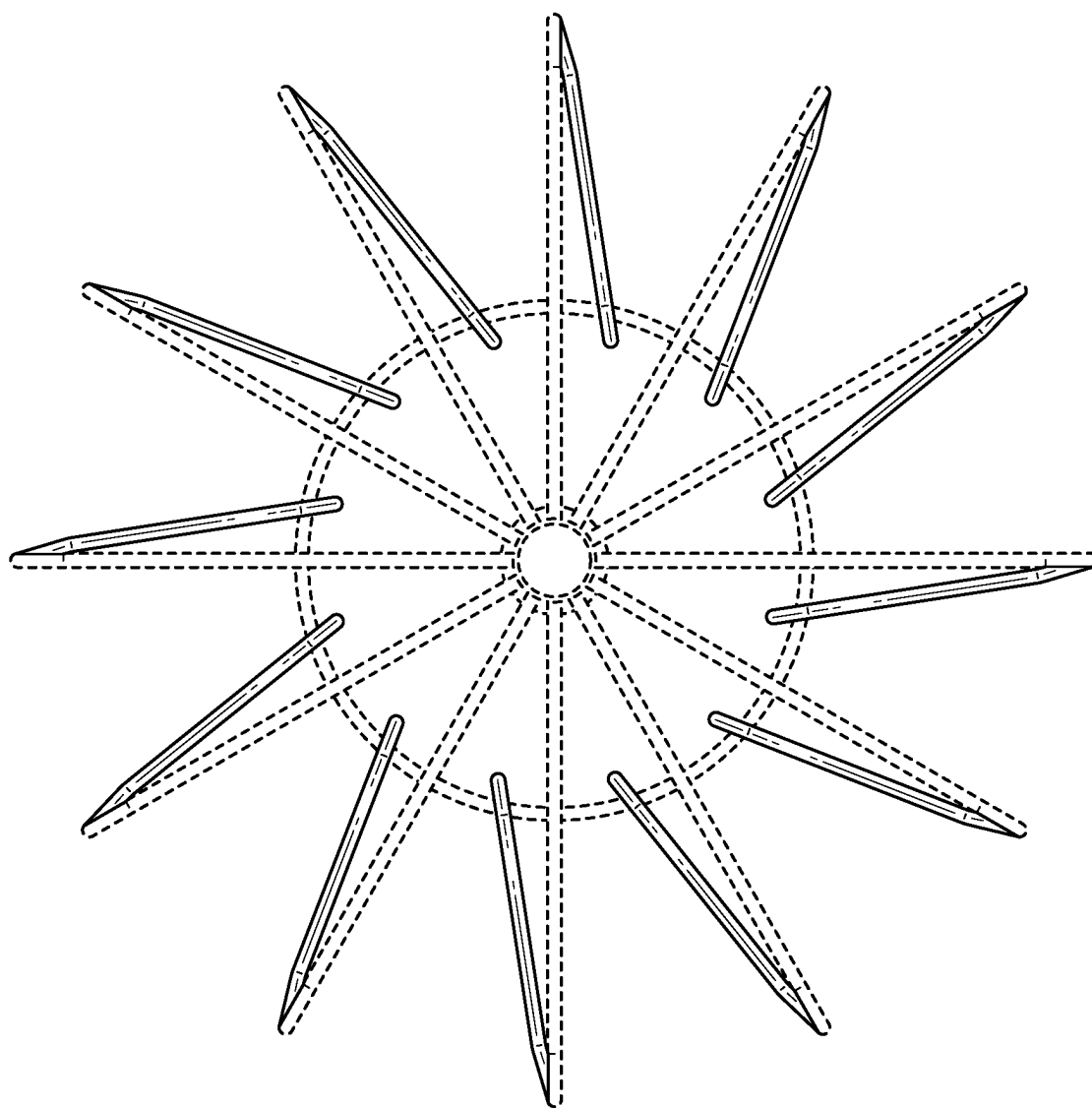


FIG. 21

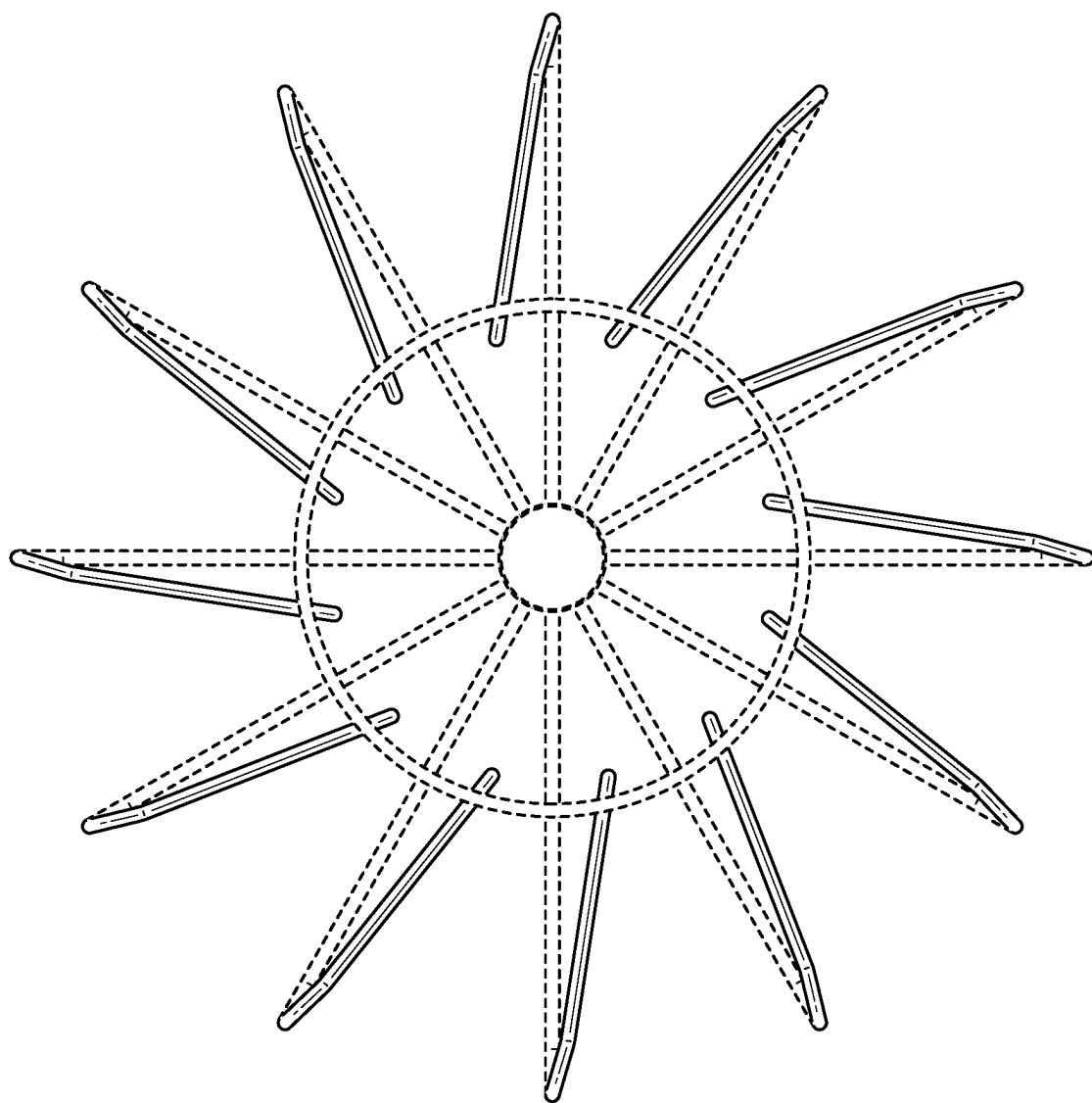


FIG. 22

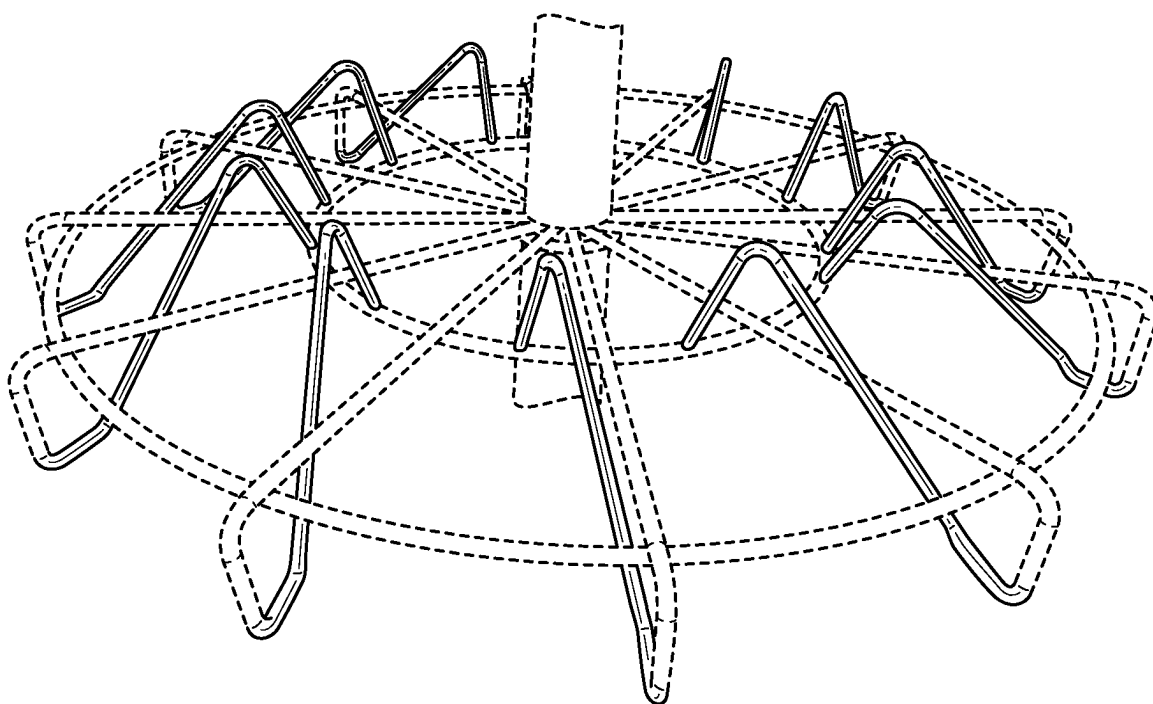


FIG. 23

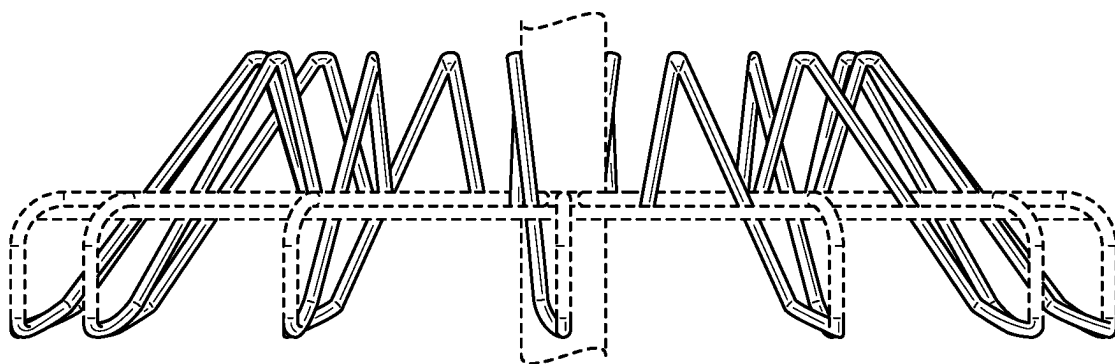


FIG. 24

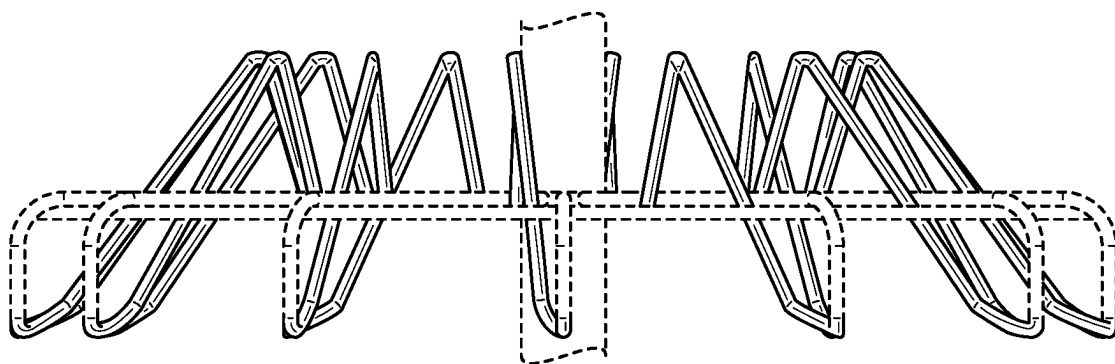


FIG. 25

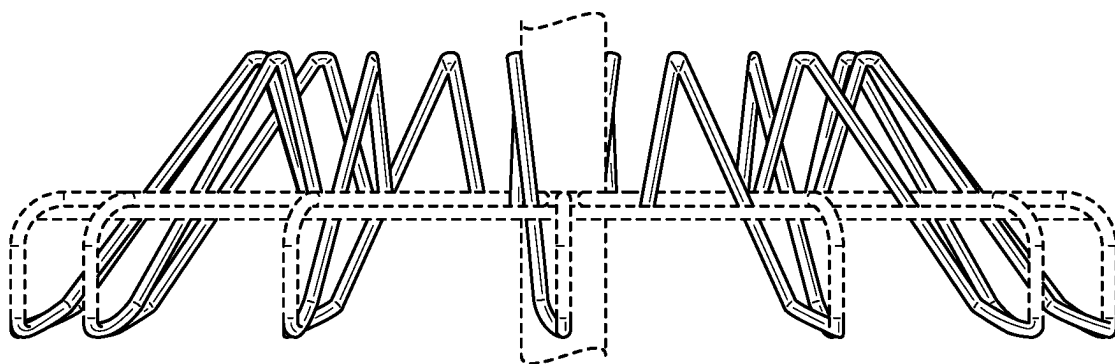


FIG. 26

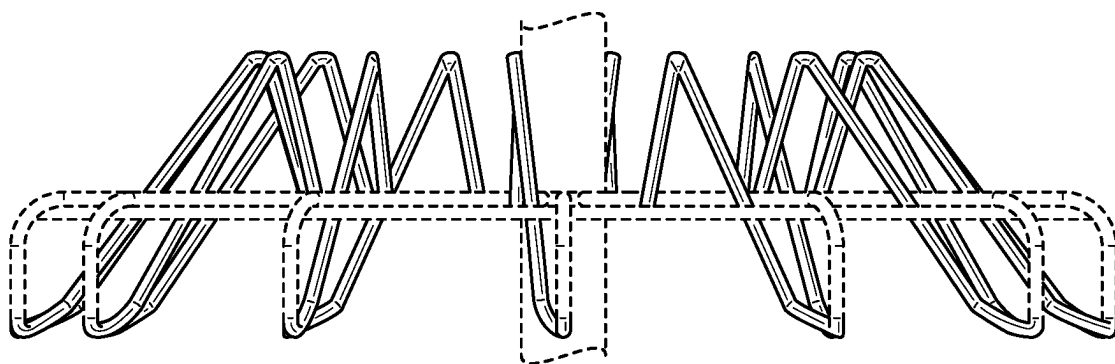


FIG. 27

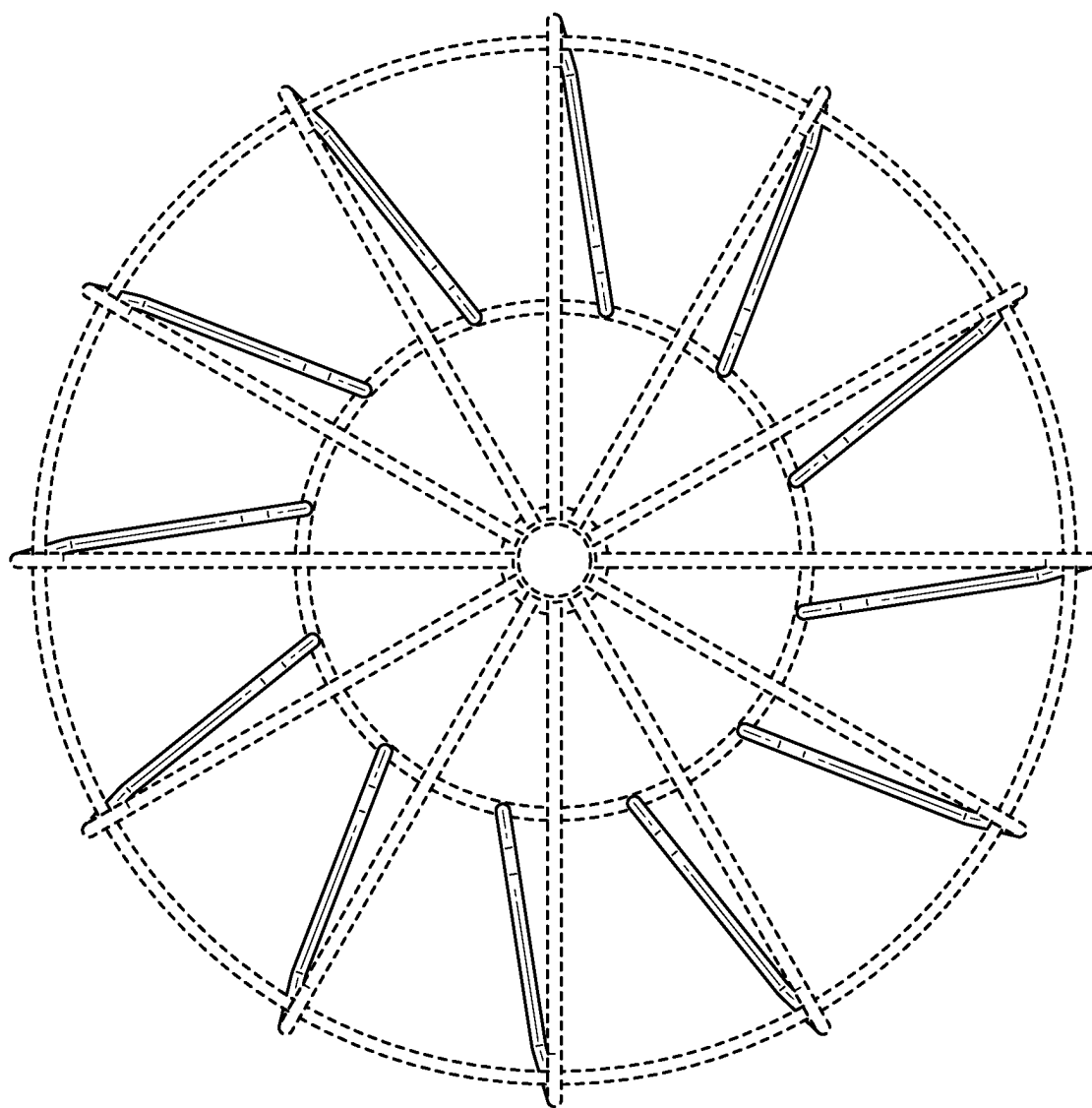


FIG. 28

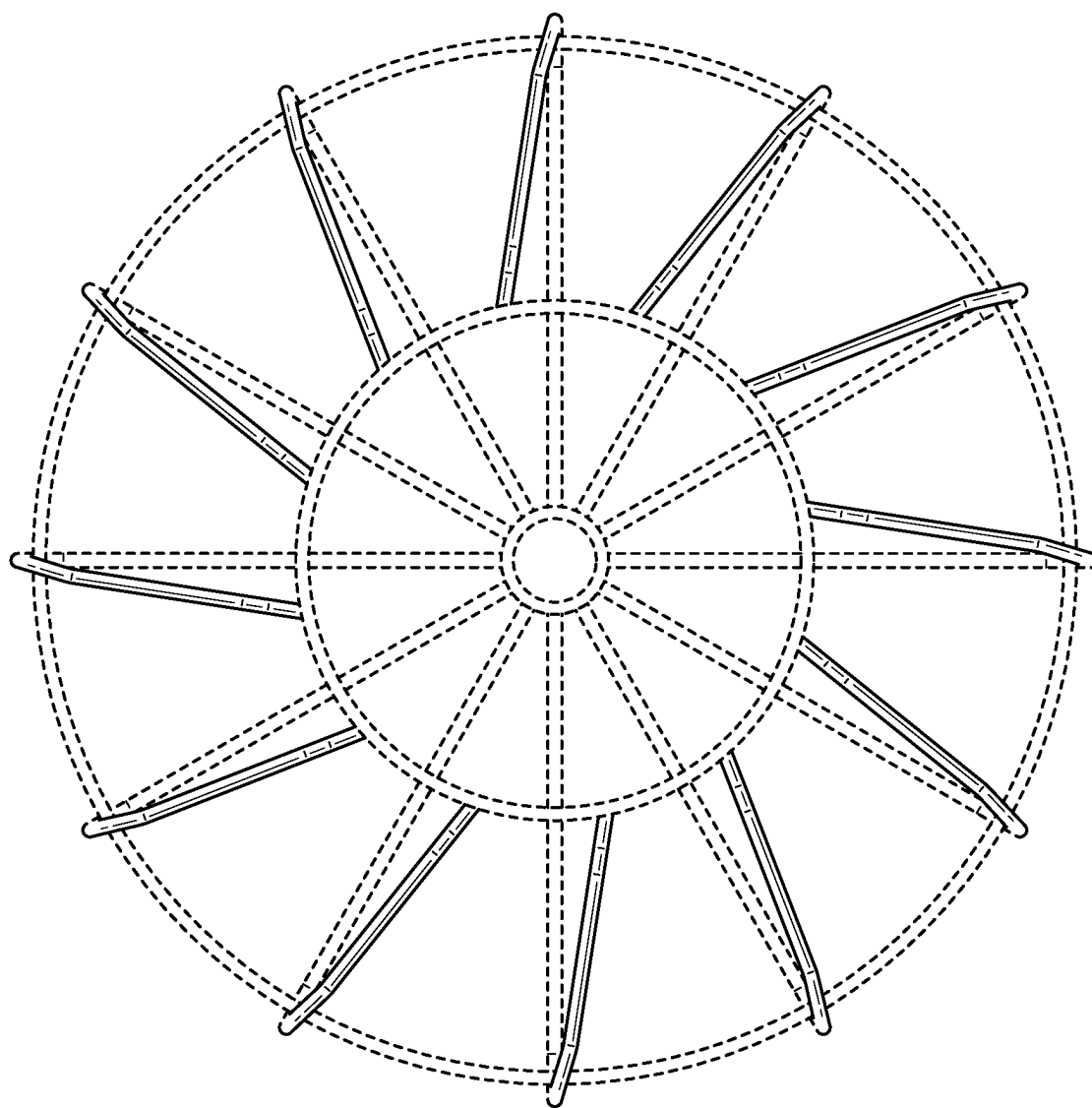


FIG. 29

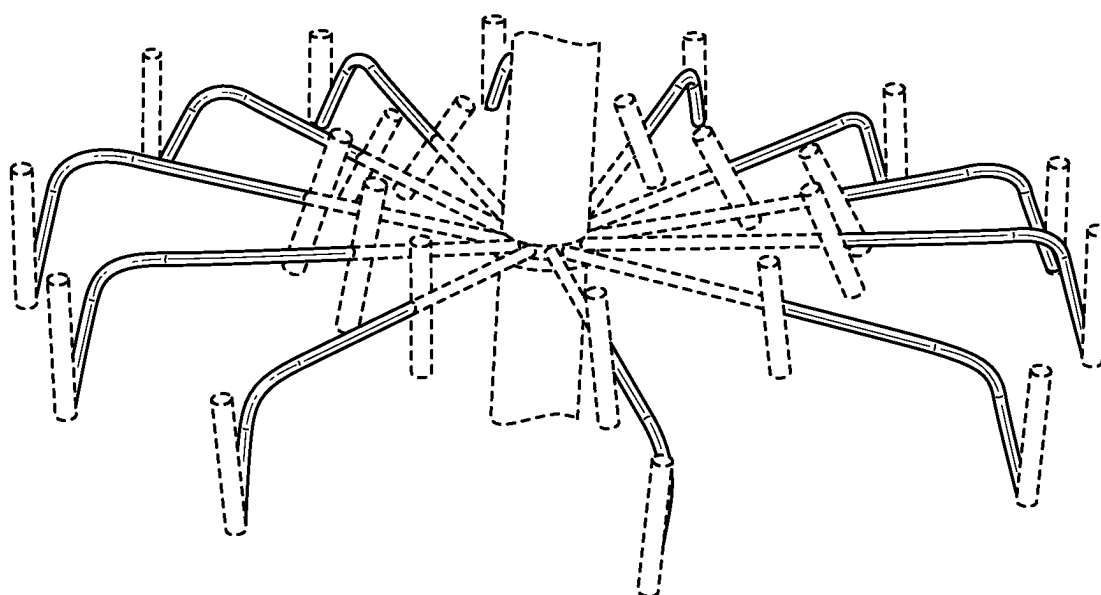


FIG. 30

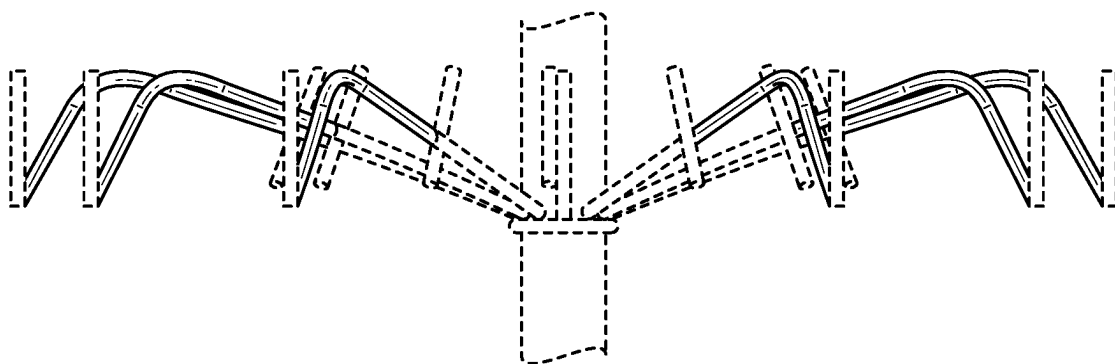


FIG. 31

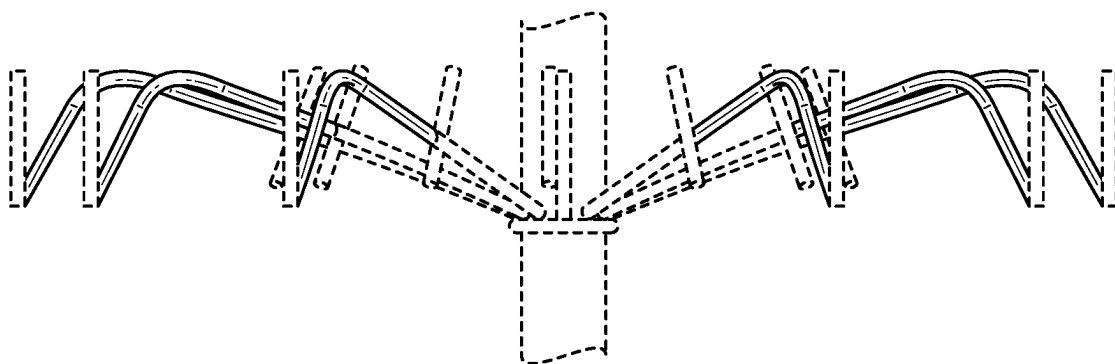


FIG. 32

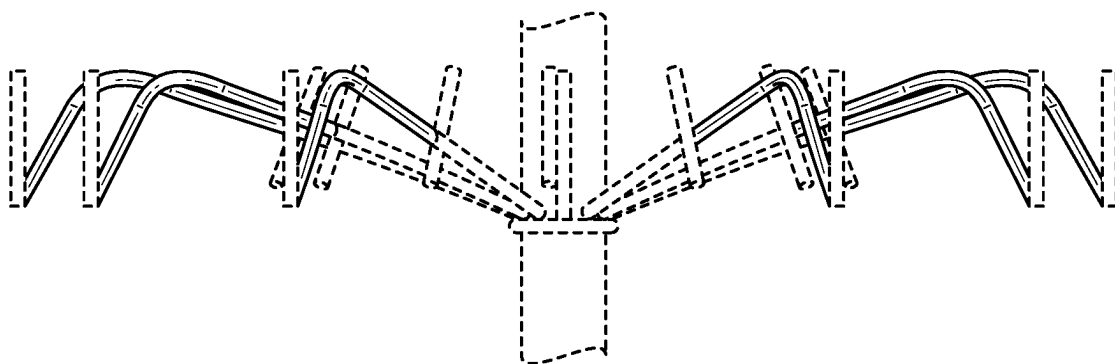


FIG. 33

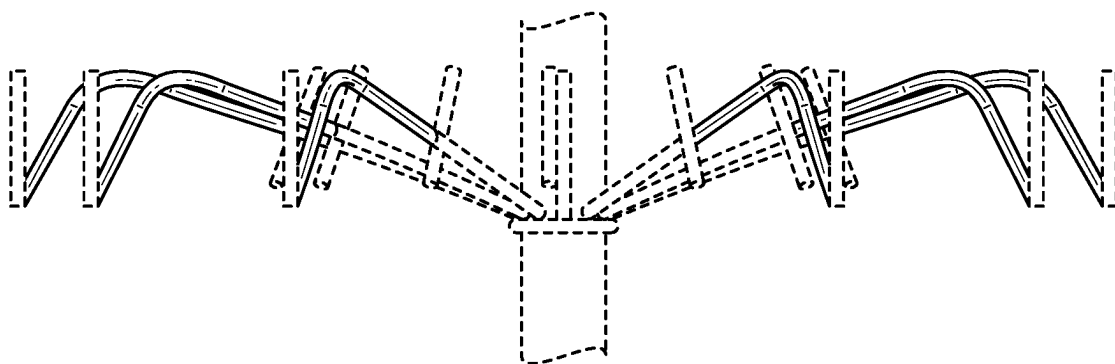


FIG. 34

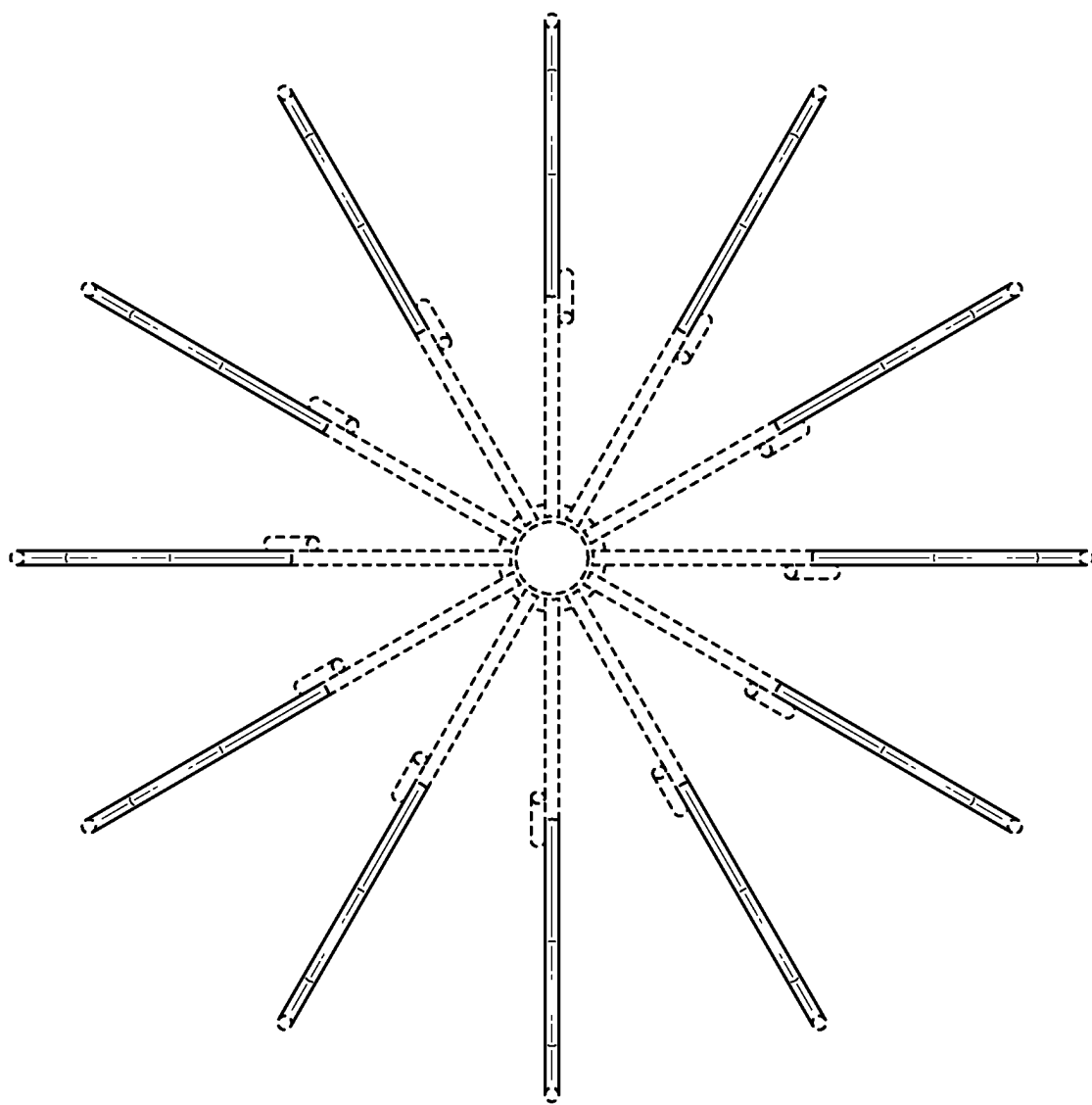


FIG. 35

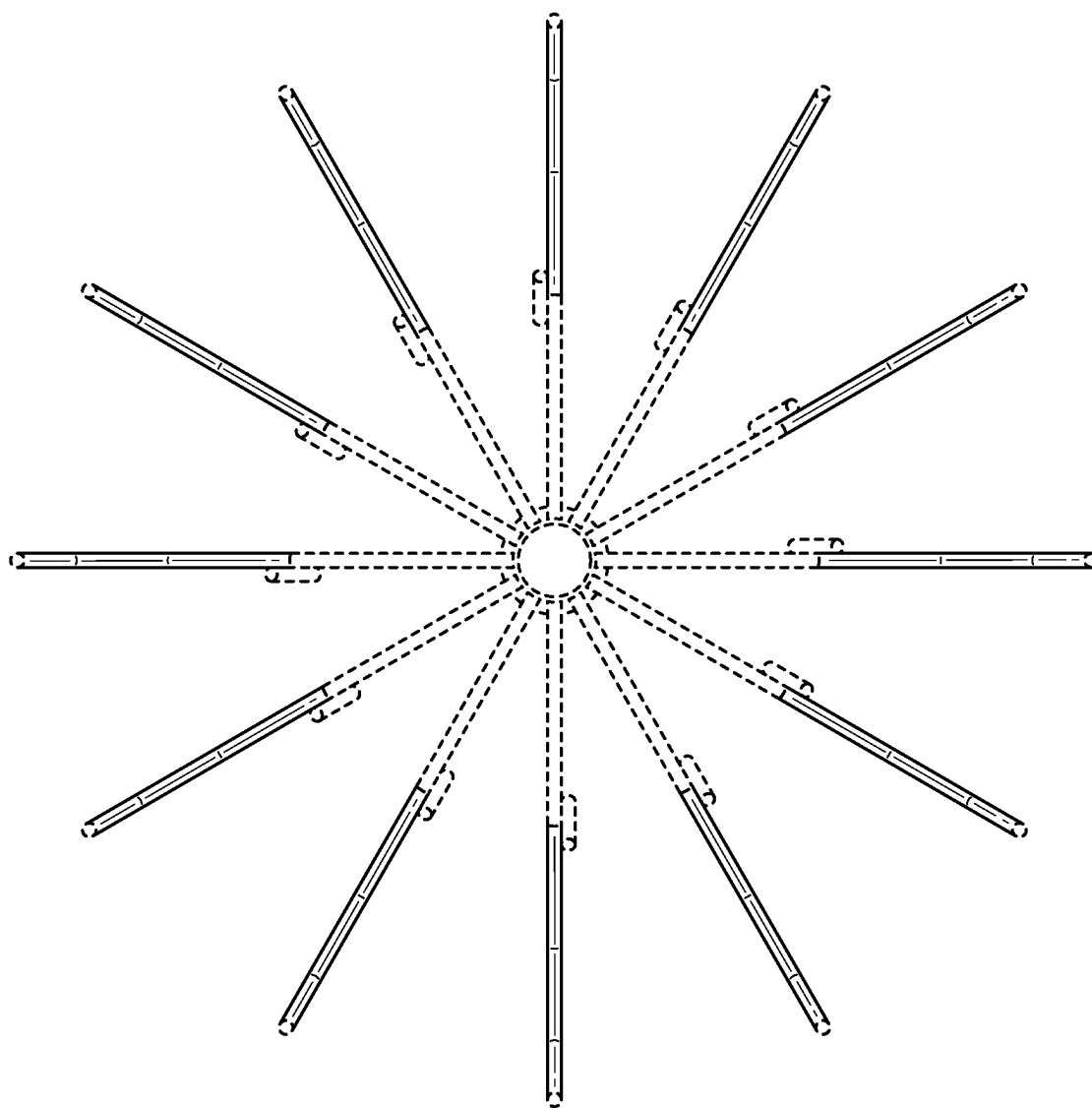


FIG. 36

SUPPORT FRAME OF AN ADJUSTABLE TARGET DISC-ENTRAPMENT DEVICE

FIELD

[0001] An adjustable target disc-catching device allowing a player to manually slide a plurality of chains from outer chain wells to inner chain wells (or vice versa) allowing the player to easily contract or expand the target of the disc-catching device, thereby making a throw-in harder (due to the smaller area of the target) or easier (due to the larger area of the target).

BACKGROUND

[0002] Disc golf is a sporting activity rapidly gaining in popularity. The game of disc golf is similar to ball golf. However, instead of hitting a golf ball with a club to direct the ball toward a given hole that catches the ball, a disc golf player throws a disc at a target that catches or entraps the disc. Generally, a disc is thrown at chains suspended above a basket. The disc generally hits the chains, slowing down the disc, then the disc falls down into the basket. When the disc comes to rest in the basket (or chains), the score is finalized. Disc golf baskets can be found on disc golf courses and are often owned by disc golf enthusiasts for personal practice.

SUMMARY

[0003] An adjustable target disc-catching device allowing a player to manually slide a plurality of chains from outer chain wells to inner chain wells (or vice versa) allowing the player to easily contract or expand the target of the disc-catching device, thereby making a throw-in harder (due to the smaller area of the target) or easier (due to the larger area of the target). For example, the standard maximum width of the target chains of a disc-catching device is about two feet across (see for example FIG. 8A). If a player wanted to become a more accurate putter (short distance throw-ins to the basket), the player could, using a device disclosed herein, easily modify the width of the target chains to be, for example, about a foot across (see for example FIG. 8B). The player could then practice putting on this smaller (and more difficult) target, increasing the players putting accuracy, translating to lower (better) scores on the disc golf course.

[0004] In one embodiment, an adjustable target disc-catching device, including:

- [0005]** a pole;
- [0006]** an upward facing basket fixed to the pole;
- [0007]** a support frame fixed to the pole above the basket, wherein the support frame includes a central pole-hub,
- [0008]** a plurality of adjustable chain holders dispersed substantially radially on the support frame;
- [0009]** wherein the adjustable chain holder includes:
 - [0010]** an outer chain well;
 - [0011]** an inner chain well; and
- [0012]** a chain hanging from the adjustable chain holder.

[0013] In one embodiment, an adjustable target disc-catching device includes a chain hanging from an adjustable chain holder, wherein the adjustable chain holder includes:

- [0014]** an outer chain well, and
- [0015]** an inner chain well;

[0016] such that the chain can reversibly be manually slid from the outer chain well to the inner chain well (and vice versa).

[0017] In one embodiment, an adjustable target disc-catching device, including a chain hanging from an adjustable chain holder, wherein the adjustable chain holder includes:

- [0018]** an outer chain barrier;
- [0019]** an inner chain barrier; and
- [0020]** a hit-sliding barrier between the outer chain barrier and the inner chain barrier; such that the chain can reversibly be manually slid from a position bounded by the outer chain barrier and the apex of the hit-sliding barrier to a position bounded by the inner chain barrier and the apex of the hit-sliding barrier, or vice versa.

[0021] In one embodiment, an adjustable target disc-catching device, including a chain hanging from an adjustable chain holder, wherein the adjustable chain holder is made of a length of solid material including:

- [0022]** an outer portion bounded by an outer chain barrier;
- [0023]** an inner portion bounded by an inner chain barrier; and
- [0024]** a hit-sliding barrier positioned between the outer portion and the inner portion.

[0025] In one embodiment, an adjustable target disc-catching device, including:

- [0026]** a pole;
- [0027]** an upward facing basket fixed to the pole;
- [0028]** a plurality of adjustable chain holders dispersed substantially radially above the basket;
- [0029]** wherein the adjustable chain holders includes:
 - [0030]** an outer chain well;
 - [0031]** an inner chain well; and
- [0032]** a chain hanging from the adjustable chain holder.

[0033] In one embodiment, an adjustable target disc-catching device, including a chain hanging from a rail system, wherein the rail system is a length of solid material featuring an outer end and an inner end, wherein the outer end of the rail system is attached to an outer chain barrier; the inner end of the rail system is attached to an inner chain barrier; and a hit-sliding barrier is positioned between the outer chain barrier and the inner chain barrier; such that the chain can reversibly be manually slid along the rail system from the outer chain well to the inner chain well, or from the inner chain well to the outer chain well, while not allowing a normally thrown disc from moving the chains from the outer chain well to the inner chain well (or vice versa).

[0034] In one embodiment, an adjustable target disc-catching device, including a rail, wherein the rail is an unobstructed length of solid material featuring an outer portion including an outer end and an inner portion including an inner end, wherein the outer end of the rail is attached to an outer chain barrier; the inner end of the rail is attached to an inner chain barrier; wherein the outer portion of the rail is an inclining sliding plane and the inner portion of the rail is a declining sliding plane. In one embodiment, the rail has a chain attached, such that the chain can reversibly be manually slid along the rail from the outer end of the rail to the inner end of the rail, while not allowing a normally thrown disc from moving the chain from the outer end of the rail to the inner end of the rail (or vice versa). In one

embodiment, an adjustable target disc-catching device features a plurality of rails. In one embodiment, the inclining sliding plane of the rail has an angle and length of: i) at least about 30 degrees and at least about 15 cm; ii) at least about 40 degrees and at least about 10 cm; iii) at least about 50 degrees and at least about 8 cm; iv) at least about 60 degrees and at least about 7 cm; or v) at least about 70 degrees and at least about 6 cm.

[0035] In one embodiment, an adjustable target disc-catching device, including a plurality of evenly distributed rails, wherein each rail is an unobstructed length of solid material featuring an outer portion with an outer end, an inner portion with an inner end, wherein the outer end of the rail is attached to an outer chain barrier; the inner end of the rail is attached to an inner chain barrier; wherein the outer portion of the rail is an inclining sliding plane, and the inner portion of the rail is a declining sliding plane.

[0036] In one embodiment, a support frame for an adjustable target disc-catching device, including a central pole-hub, a plurality of spokes radiating from the central pole-hub, a plurality of evenly distributed rails connecting to the spokes, wherein each rail is a length of solid material featuring an outer portion with an outer end, an inner portion with an inner end, wherein the outer end of the rail is attached to an outer chain barrier; the inner end of the rail is attached to an inner chain barrier; wherein the outer portion of the rail is an inclining sliding plane, and the inner portion of the rail is a declining sliding plane. In one embodiment, the inclining sliding plane of the rail has an angle and length of: i) at least about 30 degrees and at least about 15 cm; ii) at least about 40 degrees and at least about 10 cm; iii) at least about 50 degrees and at least about 8 cm; iv) at least about 60 degrees and at least about 7 cm; or v) at least about 70 degrees and at least about 6 cm.

[0037] In one embodiment, a rail for an adjustable target disc-catching device, including an uninterrupted length of solid material featuring an outer portion with an outer end, an inner portion with an inner end, wherein the outer end of the rail is attached to an outer chain barrier; the inner end of the rail is attached to an inner chain barrier; wherein the outer portion of the rail is an inclining sliding plane, and the inner portion of the rail is a declining sliding plane.

BRIEF DESCRIPTION OF THE DRAWINGS

[0038] FIG. 1 provides a representative depiction of an adjustable target disc-catching device.

[0039] FIG. 2A, FIG. 2B, FIG. 2C, FIG. 2D, FIG. 2E, FIG. 2F, FIG. 2G, FIG. 2H, FIG. 2I, FIG. 2J, FIG. 2K, FIG. 2L, FIG. 2M, FIG. 2N, FIG. 2O, and FIG. 2P provide various representative examples of adjustable chain holders (and rail systems).

[0040] FIG. 3A and FIG. 3B both depict representative examples of an adjustable chain holder.

[0041] FIG. 4 is a representative depiction of an adjustable target disc-catching device.

[0042] FIG. 5A is a perspective depiction of a representative support frame with spokes, an inner ring, an outer ring, and adjustable chain holders.

[0043] FIG. 5B is an overhead depiction of a representative support frame with spokes, an inner ring, an outer ring, and adjustable chain holders.

[0044] FIG. 6 depicts a representative adjustable chain holder.

[0045] FIG. 7 provides a depiction of a representative adjustable target disc-catching device.

[0046] FIG. 8A provides a depiction of a representative adjustable target disc-catching device with chains hanging from the outer chain wells; FIG. 8B provides a depiction of a representative adjustable target disc-catching device with chains hanging from the inner chain wells.

[0047] FIG. 9 is a top perspective view of a stand for a disc golf basket showing my new design.

[0048] FIG. 10 is a front view of a stand for a disc golf basket showing my new design.

[0049] FIG. 11 is a rear view of a stand for a disc golf basket showing my new design.

[0050] FIG. 12 is a left view of a stand for a disc golf basket showing my new design.

[0051] FIG. 13 is a right view of a stand for a disc golf basket showing my new design.

[0052] FIG. 14 is a top view of a stand for a disc golf basket showing my new design.

[0053] FIG. 15 is a bottom view of a stand for a disc golf basket showing my new design.

[0054] FIG. 16 is a top perspective view of a chain-support frame of an adjustable target disc golf basket showing my new design.

[0055] FIG. 17 is a front view of a chain-support frame of an adjustable target disc golf basket.

[0056] FIG. 18 is a rear view of a chain-support frame of an adjustable target disc golf basket.

[0057] FIG. 19 is a right view of a chain-support frame of an adjustable target disc golf basket.

[0058] FIG. 20 is a left view of a chain-support frame of an adjustable target disc golf basket.

[0059] FIG. 21 is a top view of a chain-support frame of an adjustable target disc golf basket.

[0060] FIG. 22 is a bottom view of a chain-support frame of an adjustable target disc golf basket.

[0061] FIG. 23 is a top perspective view of a chain-support frame of an adjustable target disc golf basket showing my new design.

[0062] FIG. 24 is a front view of a chain-support frame of an adjustable target disc golf basket.

[0063] FIG. 25 is a rear view of a chain-support frame of an adjustable target disc golf basket.

[0064] FIG. 26 is a right view of a chain-support frame of an adjustable target disc golf basket.

[0065] FIG. 27 is a left view of a chain-support frame of an adjustable target disc golf basket.

[0066] FIG. 28 is a top view of a chain-support frame of an adjustable target disc golf basket.

[0067] FIG. 29 is a bottom view of a chain-support frame of an adjustable target disc golf basket.

[0068] FIG. 30 is a top perspective view of a chain-support frame of an adjustable target disc golf basket showing my new design.

[0069] FIG. 31 is a front view of a chain-support frame of an adjustable target disc golf basket.

[0070] FIG. 32 is a rear view of a chain-support frame of an adjustable target disc golf basket.

[0071] FIG. 33 is a right view of a chain-support frame of an adjustable target disc golf basket.

[0072] FIG. 34 is a left view of a chain-support frame of an adjustable target disc golf basket.

[0073] FIG. 35 is a top view of a chain-support frame of an adjustable target disc golf basket.

[0074] FIG. 36 is a bottom view of a chain-support frame of an adjustable target disc golf basket.

DEFINITIONS

[0075] “Properly installed disc golf basket” means a substantially upright disc golf basket, wherein the pole is affixed approximately vertically to the ground. A majority of this application assumes a properly installed disc golf basket as the pull of gravity is often necessary for the chains to properly function.

[0076] “Incline” and “inclining” and “decline” and “declining” refers to a disc’s point of view going towards the pole. Incline indicates moving away from the ground in a properly installed disc golf basket (going from low to high). Decline indicates moving towards the ground in a properly installed disc golf basket (going from high to low).

[0077] “Sliding plane member” or an “inclining sliding plane” or an “inclining sliding plane member” is an incline (or ramp) upon which the chains slide after chains are struck by a thrown disc. As the “caught” (by the chains) disc falls into the basket, the chains slide down the sliding plane member, returning to their original position. Sliding plane members are generally described in U.S. Pat. No. 5,868,395, which is hereby incorporated by reference, including all references therein. Additionally, U.S. Pat. Nos. 4,039,189, 4,461,484, 4,792,143, 5,358,255, 5,452,903, 5,921,551, and 6,776,417, the entirety of each of which are incorporated herein by reference, as are all references disclosed therein. A “declining sliding plane” features a drop from the apex of the inclining ramp to the inner chain barrier. Generally, when sliding chains along a rail from the outer chain well to the inner chain well, one first pushes the chains up the inclining sliding ramp and slightly past the high point (apex) of the inclining ramp, and the chains will slide down the declining sliding plane (or drop), coming to rest in the inner chain well. A sliding plane may be straight, curved, or feature both. If curved, an angle and length can be determined from the apex of the sliding plane and the connection point of the chain barrier.

[0078] A “hit-sliding barrier” is a barrier sufficient to prevent a normally thrown disc from causing the chains to move from the outer chain well to the inner chain well, or vice versa (or from an outer chain well to an intermediate chain well, etc.). However, an operator can manually slide chains from the outer chain well to the inner chain well (or vice versa) passing the chains over the hit-sliding barrier (that is, without disconnecting the chains from the adjustable chain holder). A hit-sliding barrier may be a well-wall, a sufficiently long and/or steep inclining sliding plane, or a combination of both.

[0079] A “well-wall” is a substantially vertical, horizontal, and/or concave hit-sliding barrier. That is, from the point of view of a chain in the outer chain well looking towards the pole, a well-wall may be a substantially vertical, horizontal, and/or concave hit-sliding barrier. In one embodiment, the height of a vertical, horizontal, or concave hit-sliding barrier is at least greater than half the height of the thickness of the chain. For example, for a chain with thickness of 4 mm, the height of the hit-sliding barrier should be at least 2 mm. In one embodiment, the height of a vertical, horizontal, or concave hit-sliding barrier is at least greater than the height of the thickness of the chain. In one embodiment, the well-wall is a corkscrew shape. FIG. 2A, FIG. 2B, FIG. 2C,

FIG. 2D, FIG. 2E, FIG. 2H, FIG. 2I, FIG. 2K, FIG. 2N, and FIG. 2O feature non-exhaustive depictions of well-walls.

[0080] A hit-sliding barrier may also be an inclining sliding plane of sufficient material, and/or length, and/or incline angle such that a normally thrown disc will not cause the chains to move from the outer chain well to the inner chain well, or vice versa. A non-exhaustive list of suitable materials are rubber and metals including iron, steel, and aluminum (and can be painted or not). FIG. 2A, FIG. 2B, FIG. 2C, FIG. 2E, FIG. 2F, FIG. 2G, FIG. 2H, FIG. 2I, FIG. 2J, FIG. 2K, FIG. 2L, FIG. 2M, FIG. 2N, and FIG. 2P depict inclining sliding planes. FIG. 2A, FIG. 2B, FIG. 2C, FIG. 2E, FIG. 2H, FIG. 2I, FIG. 2K, and FIG. 2N depict adjustable chain holders with both a well-wall and an inclining sliding plane.

[0081] Without being limited by these examples, an adjustable chain holder made of steel, wherein the inner chain barrier and outer chain barrier are on approximately the same plane and are approximately 11.5 cm apart, can have a declining portion (attached to the inner chain barrier) of approximately 8 cm in length and an angle of approximately 45 degrees adjoined to an inclining portion (attached to the inner chain barrier) of approximately 8 cm in length and an angle of approximately 45 degrees (see FIG. 2J).

[0082] In general a hit-sliding barrier that is an inclining sliding plane, the steeper the incline angle, the less length needed for the inclining sliding plane. For example, an 8 cm long inclining sliding plane at 45 degrees may work as well as a 5 cm long inclining sliding plane at 70 degrees.

[0083] One skilled in the art realizes there are many different hit-sliding barriers and without undue experimentation, one skilled in the art can arrive at a sufficient set of parameters to form a hit-sliding barrier. For example, to determine if an adjustable chain holder is suitable (the hit-sliding barrier will prevent a hit chain in the outer chain well from leaving the outer chain well), one can make a prototype adjustable chain holder, hang a suitable chain from the outer chain well of the prototype adjustable chain holder, and throw a disc at the prototype adjustable chain holder. If the chain does not move from the outer chain well into the inner chain well (or an intermediate chain well), the prototype adjustable chain holder is suitable. Some lengths of suitable inclining sliding planes and their angles are shown below in Table 1. These were determined using an approximately 150 g chain hung from the outer chain well of various steel rails (of varying lengths and angles of the inclining sliding plane), with a disc thrown at the hanging chain at approximately 30 miles per hour, a typical putting speed. These were confirmed using chains hanging from support frames of the type depicted in FIG. 16 and FIG. 23, both of which have proven to keep chains in the outer chain well when a disc is thrown, but which feature a rail system that easily allows one to slide chains from the outer chain well to the inner chain well.

TABLE 1

Angles and lengths of inclining sliding planes sufficient to act as a hit-sliding barrier.	
Angle (degrees)	Length (cm)
30	15
40	10

TABLE 1-continued

Angles and lengths of inclining sliding planes sufficient to act as a hit-sliding barrier.	
Angle (degrees)	Length (cm)
50	8
60	7
70	6

Note: For a given angle, lengths may differ (for example be shorter) and still serve as an adequate hit-sliding barrier depending on a variety of factors including chain weight and friction (between chain and inclining sliding plane), effective weight of the pole-ring (the pole-ring and the chains attached to it provide a downward pull), the slack of the chain (a looser chain may progress higher up an inclining sliding plane when hit by a disc compared to a tighter chain), and the density of chains (the more chains a disc hits, the transfer of momentum will be spread out and each individual chain will progress up the inclining sliding plane less compared to the scenario where a disc hit fewer chains). With the disclosure herein, one skilled in the art would be able to use reasonable trial and error to arrive at a suitable set of parameters for an inclining sliding plane taking into consideration the foregoing.

[0084] One skilled in the art realizes that the parameters of a suitable adjustable chain holder depends on several factors, including the weight of the chain (and the effective weight when taking into consideration the pole-ring weight), friction of the materials, slack, and density of chains. For example, lighter chains, having less mass (and thus less stationary inertia), may require a longer or steeper incline. Chains or outer chain wells may be made of different materials that may affect the length and steepness of a hit-sliding barrier. For example, grippier surface (be it on the adjustable chain holder or the chain) may allow a shorter or less steep incline to be achieved. Rubber, certain paints, or other materials may be used to create a grippy surface. Further, one skilled in the art realizes that if a chain is attached to a pole-ring, it will be more difficult for a chain in an outer chain well, once hit, to slide into the inner chain well due to the greater effective weight of the chain. Also, the slackness of a chain may be a relevant parameter. For example, a “slack” chain may progress higher up an inclining sliding plane when hit by a disc compared to a “tighter” chain. That is, a slack chain that is connected on one end to the outer chain well and on the other end to the pole-ring may progress higher up an inclining sliding plane when hit by a disc compared to a chain with less slack.

[0085] “Spoke” is a length of material that connects to the central pole-hub (or to the pole itself) and radiates out. It may connect to an inner ring and/or an outer ring. In some embodiments, the material that constitutes the spoke (or a portion thereof) may serve as the adjustable chain holder if the spoke features a chain barrier. For example, FIG. 7 depicts a spoke radiating from the pole. The same spoke material extends beyond element 7-2 (which serves as an inner chain barrier). A spoke may be straight, curved, or bent.

[0086] A “chain barrier” is a piece of material bounding the chain well (or the rail system) that keeps the chains hanging from the chain well. An “outer chain barrier” is a piece of material that can keep the chains hanging in the

outer chain well. The outer chain barrier may be the outer ring, as shown in FIG. 8A and FIG. 8B. An “inner chain barrier” is a piece of material that can keep the chains hanging from the inner chain well. The inner chain barrier may be the inner ring, as depicted in FIG. 4 (and FIG. 8A and FIG. 8B). However, the chain barrier may be a piece of material that extends from or crosses the inner end or outer end of the rail system to form a width greater than the inner diameter of the chain such that the chain cannot pass, for example as depicted by element 7-2 in FIG. 7. The chain barrier may be a piece of material that extends vertically from the inner end or outer end of the rail system, for example as depicted by element 7-1 in FIG. 7 and as depicted by element 6-2 in FIG. 6. An inner chain barrier and an outer chain barrier may separately have a length of greater than about 1 cm, greater than about 2 cm, greater than about 3 cm, greater than about 4 cm, greater than about 5 cm, greater than about 6 cm, greater than about 7 cm, greater than about 8 cm, greater than about 9 cm, or greater than about 10 cm.

[0087] A “chain well” is a portion of the adjustable chain holder (or rail system) that a chain is capable of hanging from when not in use and is also contained within while in ongoing use. By way of contrast, although it may be possible to balance a chain on a vertical hit-sliding barrier (a well-wall), ongoing use (for example, multiple thrown discs hitting the chains) will shift the chains to another position (for example a disc hitting such a chain may knock the chain into the inner chain well). An “outer chain well” is a well bounded on the outer portion by an outer chain barrier and on the inner portion by the apex of the closest hit-sliding barrier. Such an outer chain barrier may be the outer ring or another piece of material that keeps the chains hanging in the outer chain well. An outer chain well has a low point (or other minima of potential energy) from which the chains will be hanging when not in use. For example, when chains are in the outer chain well, a thrown disc may cause the chains to slide up an inclining sliding plane, which then slide back to the low point, with the chains being contained within the outer chain well during the entire described sequence of events. An “inner chain well” is a well bounded on the inner portion by an inner chain barrier and on the outer portion by the apex of the closest hit-sliding barrier. Such an inner chain barrier may be the inner ring, or another piece of material that keeps the chains hanging in the inner chain well. Similarly, an inner chain well has a low point (or other minima of potential energy) from which the chains will be hanging when not in use. The low point of a chain well from which the chains will hang when not being actively used has a length. For example, a chain hanging in the outer chain well of the adjustable chain holder depicted in FIG. 6 will hang at the juncture of element 6-2 and element 6-3 and have a low point length of about the width of the chain. However, given the flat bottom of the outer chain well of FIG. 2D, a chain can hang within the area bounded by the hit-sliding barrier and the outer chain barrier. In one embodiment a chain well has a low point length of less than about 5 mm, less than about 1 cm, less than about 1.5 cm, less than about 2 cm, less than about 2.5 cm, less than about 3 cm, less than about 3.5 cm, less than about 4 cm, less than about 4.5 cm, less than about 5 cm, less than about 5.5 cm, or about 6 cm or more. In one embodiment a chain well has a low point length of about the width of a chain. A shorter low point length allows for greater consistency of distance from the

outer (or inner) chain well to the pole. The shortest low point length can be achieved by a non-flat bottom of the chain well, as depicted for example in FIG. 2A, FIG. 2E, FIG. 2F (these are non-exhaustive examples). The juncture of an inclining sliding plane (or a declining sliding plane) and a hit-sliding barrier forms a short low point (of about the width of a chain), for example as depicted in FIG. 8A and FIG. 8B. An adjustable chain holder may have only one outer chain well and only one inner chain well. An adjustable chain holder may have more than two chain wells. For example, an adjustable chain holder may have an outer chain well, an intermediate chain well, and an inner chain well, each having a low point different distances from the pole, for example as depicted in FIG. 2P. A chain well does not necessarily rely upon gravity for the chain to sit within it. For example, a chain well may be made by magnetic attraction connecting the chain to the outer chain barrier or the inner chain barrier. Without being limited by this example, a chain hanging from the low point of the outer chain well may be approximately 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, or 32 centimeters from the pole. Without being limited by this example, a chain hanging from the low point of the inner chain well may be approximately 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, or 17 centimeters from the pole. One skilled in the art realizes these distances are customizable depending on the needs of a disc golfer.

[0088] “Stand” indicates a support for holding the pole substantially vertical. Without being limited by this example, a disc-catching device may have a substantially circular frame affixed to the bottom portion of a pole, as depicted by element 1-2 in FIG. 1. FIG. 9 depicts a stand of a disc golf basket featuring a “byrdy foot” design. A pole can connect to this stand. However, a stand may be a hole in the ground that the pole fits inside (this style is most common on disc golf courses). A stand may also be connected to the top of the pole, for example a chain hanging from a tree branch connected to the top of an adjustable target disc-catching device.

[0089] A pole-hub (or a central pole-hub) is the central part of the support frame that connects the support frame to the pole. For example, element 1-6 in FIG. 1 shows a central pole-hub. A central pole-hub can be seen at least in FIG. 4, FIG. 5A, FIG. 5B, FIG. 8A, FIG. 8B, FIG. 16, FIG. 21, FIG. 22, FIG. 23, FIG. 28, FIG. 29, FIG. 30, FIG. 31, FIG. 32, FIG. 33, FIG. 34, FIG. 35, and FIG. 36. A central pole-hub may connect to the pole with a screw. For example, the center of the central pole-hub may be a negative space, through which a screw can pass that secures the central pole-hub to the pole. However, the central pole-hub may include a screw, which can be rotated to secure the support frame to the pole.

[0090] “Reversibly manually slid” or “reversibly slid” or similar terms indicate that the chain can be manually slid back and forth from the outer chain well to the inner chain well.

[0091] “Chain hanging from” or “chains hanging from” or “chains attached to” or “chain attached to” or other such language indicates the inner portion of a chain or an attachment thereto (for example, an S-hook) is passed through by the object in question or otherwise connected. For example, a chain hanging from the adjustable chain holder may mean a chain is connected through the bottom portion of an S-hook, while the top portion of the S-hook wraps over the adjustable chain holder.

[0092] “Chain” and “chains” may include any device to attach the chain to an object (for example an S-hook). One skilled in the art understands there are many means of connecting a chain to the adjustable chain holder and/or support frame and/or pole-ring. Additionally, “chain” may encompass various modalities of slowing the momentum of a disc, including but not limited to a connected flexible series of links, cable, rope, bungee cord, a net, or other materials suitable to slow the momentum of a disc allowing the disc to drop into the basket. The top of a chain is generally attached to an adjustable chain holder or a stationary chain holder. The bottom of a chain is generally attached to a pole-ring. In some embodiments, a “support frame” is referred to as a “chain-support frame”.

[0093] “Inner” generally means closer to the pole while “outer” generally means further from the pole.

[0094] “Dispersed substantially radially” when referencing adjustable chain holders (or rail systems) indicates the adjustable chain holders (or rail systems) are arranged in a pattern and that the inner chain well is closer to the pole (or center) than the outer chain well, however the inner chain well and outer chain well are not necessarily on a straight line connecting to the pole. For example, FIG. 5B depicts adjustable chain holders that deviate by about 15 degrees from a line extending from the pole. That is, the line connecting the inner chain well and outer chain well intersects the line connecting said outer chain well and the pole at an angle of about 15 degrees. Such deviation from the line connecting the pole to the outer chain well allows for a longer inclining sliding plane and allow for easier sliding chains from the outer chain well to the inner chain well (and vice versa). In one embodiment the line connecting the inner chain well and outer chain well intersects the line connecting said outer chain well and the pole at an angle of about 0 degrees, about 1 degrees, about 2 degrees, about 3 degrees, about 4 degrees, about 5 degrees, about 6 degrees, about 7 degrees, about 8 degrees, about 9 degrees, about 10 degrees, about 11 degrees, about 12 degrees, about 13 degrees, about 14 degrees, about 15 degrees, about 16 degrees, about 17 degrees, about 18 degrees, about 19 degrees, about 20 degrees, about 25 degrees, about 30 degrees, about 35 degrees, about 40 degrees, about 45 degrees, about 50 degrees, about 55 degrees, about 60 degrees, about 65 degrees, about 70 degrees, about 75 degrees, about 80 degrees, or about 85 degrees. In some embodiments, the line connecting the inclining sliding plane intersects the line connecting the outer chain well and the pole at an angle of about 0 degrees, about 1 degrees, about 2 degrees, about 3 degrees, about 4 degrees, about 5 degrees, about 6 degrees, about 7 degrees, about 8 degrees, about 9 degrees, about 10 degrees, about 11 degrees, about 12 degrees, about 13 degrees, about 14 degrees, about 15 degrees, about 16 degrees, about 17 degrees, about 18 degrees, about 19 degrees, about 20 degrees, about 25 degrees, about 30 degrees, about 35 degrees, about 40 degrees, about 45 degrees, about 50 degrees, about 55 degrees, about 60 degrees, about 65 degrees, about 70 degrees, about 75 degrees, about 80 degrees, or about 85 degrees.

[0095] “Rail system” (or “rail”) indicates a length of material (a rail) passes through a chain such that the chain can be reversibly slid along the length of rail, for example from an outer chain well to an inner chain well. The rail system disclosed herein can have an outer end, an inner end, and a hit-sliding barrier between the outer end and the inner

end, wherein the outer end is attached to an outer chain barrier and the inner end is attached to an inner chain barrier. The rail system can be connected to a pole in various ways, including via a spoke or via a support frame. For example, referencing FIG. 7, elements 7-1 are outer chain barriers, elements 7-2 are inner chain barriers, and the length of material between element 7-1 and element 7-2 is the rail system from which a chain is hanging. The chain can be manually slid from the outer chain well to the inner chain well (or vice versa). The two rail systems depicted in FIG. 7 radiate out from the pole, connected to the pole via a spoke, which attaches to the inner end of the rail system. Referencing FIG. 4, depicted are six rail systems dispersed substantially radially above the basket, and are attached to a support frame. FIG. 16, FIG. 17, FIG. 18, FIG. 19, FIG. 20, FIG. 21, FIG. 22, FIG. 23, FIG. 24, FIG. 25, FIG. 26, FIG. 27, FIG. 28, FIG. 29, FIG. 30, FIG. 31, FIG. 32, FIG. 33, FIG. 34, FIG. 35, and FIG. 36 each show examples of a plurality of rail systems in solid lines. Purely for design patent purposes, the broken lines in the drawings are used to illustrate unclaimed portions of the chain-support frame of an adjustable target disc golf basket and form no part of the claimed design. However, the intersection of the solid portion of the drawings (showing the rail system) with the broken lines shows the rail system connecting to outer chain barriers and inner chain barriers. Chains are not shown hanging from these figures for clarity, however the shown rail systems can pass through a chain, such that the chain can be reversibly slid along the length of rail, for example from an outer chain well to an inner chain well.

[0096] A “support frame” (or “chain-support frame”) is the unit structure from which the chains hang, which connects to the pole. The support frame may include spokes, rails, inner chain barriers (which may be inner ring), outer chain barriers (which may be outer ring), and a central pole-hub. FIG. 5A, FIG. 5B, FIG. 16, FIG. 17, FIG. 18, FIG. 19, FIG. 20, FIG. 21, FIG. 22, FIG. 23, FIG. 24, FIG. 25, FIG. 26, FIG. 27, FIG. 28, FIG. 29, FIG. 30, FIG. 31, FIG. 32, FIG. 33, FIG. 34, FIG. 35, and FIG. 36, show a support frame with no chains hanging from the chain wells.

[0097] “Uninterrupted” when describing a rail indicates that there is no material on the length of solid material that forms the rail which could prevent a chain on the rail from being slid from the outer chain well to the inner chain well (or vice versa). That is, along the rail between the outer chain barrier and the inner chain barrier, there is no other chain barrier.

[0098] “Substantially” and “about” and “approximately” mean within 25%.

DETAILED DESCRIPTION

[0099] In one embodiment, an adjustable target disc-catching device, including:

- [0100]** a pole;
- [0101]** an upward facing basket fixed to the pole;
- [0102]** a support frame fixed to the pole above the basket, wherein the support frame includes:
 - [0103]** a central pole-hub,
 - [0104]** a plurality of spokes extending radially from the central pole-hub,
 - [0105]** an outer ring connected to the spokes;
- [0106]** a plurality of adjustable chain holders dispersed radially on the support frame;

[0107] a chain hanging from the adjustable chain holder,

[0108] wherein the adjustable chain holder includes:

[0109] an outer chain well, and

[0110] an inner chain well,

[0111] wherein the chain hanging from the adjustable chain holder can be reversibly manually slid from the outer chain well to the inner chain well, or from the inner chain well to the outer chain wall.

[0112] In one embodiment, the support frame further includes an inner ring.

[0113] In one embodiment, the inner ring includes a stationary chain support.

[0114] In one embodiment, the adjustable chain holders are evenly dispersed substantially radially on the support frame.

[0115] In one embodiment, the adjustable target disc-catching device further includes a pole-ring. In one embodiment, the bottom portion of the chains connect to the pole-ring.

[0116] In one embodiment, the outer chain well includes a inclining sliding plane having an incline such that a disc may strike a chain and have energy absorbed thereby by the travel of the top end of the chain up the inclining sliding plane, and the disc drops into the basket, and the chain will return to its original position by gravity.

[0117] In one embodiment, the inclining sliding plane has an incline of about thirty degrees.

[0118] In one embodiment, the inclining sliding plane has a length of about two inches.

[0119] In one embodiment, the adjustable chain holder includes a hit-sliding barrier between the outer chain well and the inner chain well.

[0120] In one embodiment, the hit-sliding barrier is substantially vertical.

[0121] In one embodiment, the hit-sliding barrier is substantially horizontal.

[0122] In one embodiment, the hit-sliding barrier is a cork-screw shape.

[0123] In one embodiment, a stand is affixed to the pole.

[0124] In one embodiment, a stand is affixed to the bottom of the pole.

[0125] In one embodiment, a stand is affixed to the top of the pole.

[0126] In one embodiment, the adjustable chain holder is a portion of the spoke.

[0127] In one embodiment, the support frame includes an outer ring.

[0128] In one embodiment, the support frame includes a plurality of spokes dispersed substantially radially on the support frame.

[0129] In one embodiment, the outer chain well includes an inclining sliding plane.

[0130] In one embodiment, the hit-sliding barrier is an inclining sliding plane.

[0131] In one embodiment, the inclining sliding plane is at least about 30 degrees, at least about 40 degrees, at least about 45 degrees, at least about 50 degrees, at least about 60 degrees, or at least about 75 degrees.

[0132] In one embodiment, the hit-sliding barrier is a well-wall.

[0133] In one embodiment, the well-wall is substantially vertical.

- [0134] In one embodiment, the well-wall is substantially concave.
- [0135] In one embodiment, the well-wall is substantially horizontal.
- [0136] In one embodiment, the well-wall is partially vertical.
- [0137] In one embodiment, the well-wall is partially concave.
- [0138] In one embodiment, the well-wall is partially horizontal.
- [0139] In one embodiment, the well-wall is approximately vertical.
- [0140] In one embodiment, the well-wall is approximately concave.
- [0141] In one embodiment, the well-wall is approximately horizontal.
- [0142] In one embodiment, the adjustable chain holder is a portion of a spoke.
- [0143] In one embodiment, the pole is height-adjustable, allowing for altering the height of the basket and chains.
- [0144] In one embodiment, the low point of the outer chain well is about 5 mm long.
- [0145] In one embodiment, the low point of the outer chain well is about 1 cm long.
- [0146] In one embodiment, the low point of the outer chain well is about 1.5 cm long.
- [0147] In one embodiment, the low point of the outer chain well is about 2 cm long.
- [0148] In one embodiment, the low point of the outer chain well is about 2.5 cm long.
- [0149] In one embodiment, the low point of the outer chain well is about 3 cm long.
- [0150] In one embodiment, an adjustable target disc-catching device includes a chain hanging from an adjustable chain holder, wherein the adjustable chain holder includes:
- [0151] an outer chain well, and
 - [0152] an inner chain well;
 - [0153] such that the chain can reversibly be manually slid from the outer chain well to the inner chain well, or from the inner chain well to the outer chain well.
- [0154] In one embodiment, the outer chain well includes an inclining sliding plane.
- [0155] In one embodiment, the hit-sliding barrier is an inclining sliding plane.
- [0156] In one embodiment, the inclining sliding plane is at least about 30 degrees, at least about 45 degrees, at least about 60 degrees, or at least about 75 degrees.
- [0157] In one embodiment, the hit-sliding barrier is a well-wall.
- [0158] In one embodiment, the well-wall is substantially vertical.
- [0159] In one embodiment, the well-wall is substantially concave.
- [0160] In one embodiment, the well-wall is substantially horizontal.
- [0161] In one embodiment, the well-wall is partially vertical.
- [0162] In one embodiment, the well-wall is partially concave.
- [0163] In one embodiment, the well-wall is partially horizontal.
- [0164] In one embodiment, the well-wall is approximately vertical.
- [0165] In one embodiment, the well-wall is approximately concave.
- [0166] In one embodiment, the well-wall is approximately horizontal.
- [0167] In one embodiment, the adjustable chain holder is a portion of a spoke.
- [0168] In one embodiment, the adjustable target disc-catching device, further includes a pole; an upward facing basket fixed to the pole; and a plurality of adjustable chain holders dispersed substantially radially above the basket.
- [0169] In one embodiment, an adjustable target disc-catching device, including a chain hanging from an adjustable chain holder, wherein the adjustable chain holder includes:
- [0170] an outer chain barrier;
 - [0171] an inner chain barrier;
 - [0172] a hit-sliding barrier positioned between the outer chain barrier and the inner chain barrier; such that the chain can reversibly be manually slid from a position bounded by the outer chain barrier and the hit-sliding barrier (outer chain well) to a position bounded by the inner chain barrier and the hit-sliding barrier (inner chain well), or from a position bounded by the inner chain barrier and the hit-sliding barrier to a position bounded by the outer chain barrier and the hit-sliding barrier.
- [0173] In one embodiment, the outer chain barrier is an outer ring.
- [0174] In one embodiment, the inner chain barrier is an inner ring.
- [0175] In one embodiment, the portion of the adjustable chain holder between the outer chain barrier and the hit-sliding barrier includes an inclining sliding plane.
- [0176] In one embodiment, the hit-sliding barrier is an inclining sliding plane.
- [0177] In one embodiment, the hit-sliding barrier that is an inclining sliding plane is at least about 30 degrees, at least about 45 degrees, at least about 60 degrees, or at least about 75 degrees.
- [0178] In one embodiment, the hit-sliding barrier is a well-wall.
- [0179] In one embodiment, the well-wall is substantially vertical.
- [0180] In one embodiment, the well-wall is substantially concave.
- [0181] In one embodiment, the well-wall is substantially horizontal.
- [0182] In one embodiment, the well-wall is partially vertical.
- [0183] In one embodiment, the well-wall is partially concave.
- [0184] In one embodiment, the well-wall is partially horizontal.
- [0185] In one embodiment, the well-wall is approximately vertical.
- [0186] In one embodiment, the well-wall is approximately concave.
- [0187] In one embodiment, the well-wall is approximately horizontal.
- [0188] In one embodiment, the adjustable chain holder is a portion of a spoke.
- [0189] In one embodiment, the adjustable target disc-catching device, further includes a pole; an upward facing basket fixed to the pole; and a plurality of adjustable chain holders dispersed substantially radially above the basket.
- [0190] In one embodiment, an adjustable target disc-catching device, including:
- [0191] a pole;
 - [0192] an upward facing basket fixed to the pole;
 - [0193] a plurality of adjustable chain holders dispersed substantially radially above the basket;

[0194] wherein the adjustable chain holders include:

[0195] an outer chain well;

[0196] an inner chain well; and

[0197] a chain hanging from the adjustable chain holder.

[0198] In one embodiment, the hit-sliding barrier is a well-wall.

[0199] In one embodiment, the hit-sliding barrier is an inclining sliding plane.

[0200] In one embodiment, the inclining sliding plane is at least about 30 degrees, at least about 45 degrees, at least about 60 degrees, or at least about 75 degrees.

[0201] In one embodiment, the adjustable target disc-catching device further includes an inner ring.

[0202] In one embodiment, the adjustable target disc-catching device further includes an outer ring.

[0203] In one embodiment, a stand is fixed to the bottom of the pole.

[0204] In one embodiment, the adjustable target disc-catching device further includes a stationary chain support. In one embodiment, a chain hangs from the stationary chain support.

[0205] In one embodiment, an adjustable target disc-catching device, including a chain hanging from an adjustable chain holder, wherein the adjustable chain holder includes a length of solid material including:

[0206] an outer portion bounded by an outer chain barrier;

[0207] an inner portion bounded by an inner chain barrier; and

[0208] a hit-sliding barrier positioned between the outer portion and the inner portion; such that the chain can reversibly be manually slid from a position bounded by the outer chain barrier and the hit-sliding barrier (outer chain well) to a position bounded by the inner chain barrier and the hit-sliding barrier (inner chain well), or from a position bounded by the inner chain barrier and the hit-sliding barrier to a position bounded by the outer chain barrier and the hit-sliding barrier.

[0209] In one embodiment, the outer chain barrier is an outer ring.

[0210] In one embodiment, the inner chain barrier is an inner ring.

[0211] In one embodiment, the hit-sliding barrier is an inclining sliding plane.

[0212] In one embodiment, the inclining sliding plane is at least about 30 degrees, at least about 45 degrees, at least about 60 degrees, or at least about 75 degrees.

[0213] In one embodiment, the hit-sliding barrier is a well-wall.

[0214] In one embodiment, the well-wall is substantially vertical.

[0215] In one embodiment, the well-wall is substantially concave.

[0216] In one embodiment, the well-wall is substantially horizontal.

[0217] In one embodiment, the adjustable chain holder further includes a chain hanging from the adjustable chain holder.

[0218] In one embodiment, the hit-sliding-barrier that is an inclining sliding plane has an incline angle of more than about 30 degrees.

[0219] In one embodiment, the hit-sliding-barrier that is an inclining sliding plane has an incline angle of more than about 45 degrees.

[0220] In one embodiment, the hit-sliding-barrier that is an inclining sliding plane has an incline angle of more than about 60 degrees.

[0221] In one embodiment, the hit-sliding-barrier that is an inclining sliding plane has an incline angle of more than about 75 degrees.

[0222] In one embodiment, the adjustable target disc-catching device, further includes a pole; an upward facing basket fixed to the pole; and a plurality of adjustable chain holders dispersed substantially radially above the basket.

[0223] In one embodiment, a method of adjusting the target area of a disc-catching device, the method including: manually sliding a plurality of chains from outer chain wells to inner chain wells, or manually sliding a plurality of chains from inner chain wells to outer chain wells.

[0224] In one embodiment, a method of adjusting the target area of a disc-catching device, the method including: manually sliding a plurality of chains along a rail system from outer chain wells to inner chain wells, or manually sliding a plurality of chains along the rail system from inner chain wells to outer chain wells.

[0225] In one embodiment, an adjustable target disc-catching device, including a chain hanging from a rail system, wherein the rail system is a length of solid material featuring an outer end and an inner end, wherein the outer end of the rail system is attached to an outer chain barrier; the inner end of the rail system is attached to an inner chain barrier; and a hit-sliding barrier is positioned between the outer chain barrier and the inner chain barrier; such that the chain can reversibly be manually slid along the rail system from a position bounded by the outer chain barrier and the apex of the hit-sliding barrier (the outer chain well) to a position bounded by the inner chain barrier and the apex of the hit-sliding barrier (the inner chain well), or from a position bounded by the inner chain barrier and the apex of the hit-sliding barrier to a position bounded by the outer chain barrier and the apex of the hit-sliding barrier (the inner chain well configured so as to not allow a normally thrown disc from moving the chains out from the outer chain well).

[0226] In one embodiment, the adjustable target disc-catching device further includes a pole; an upward facing basket fixed to the pole; and a plurality of rail systems dispersed substantially radially above the basket.

[0227] In one embodiment, an adjustable target disc-catching device, including a chain hanging from a rail system, wherein the rail system is a length of solid material featuring an outer end, an inner end, and a middle portion, wherein the outer end of the rail system is attached to an outer chain barrier; the inner end of the rail system is attached to an inner chain barrier; and the middle portion is elevated above the outer end and the inner end, such that the chain hanging from the rail system can be manually slid from the outer end to the inner end, or from the inner end to the outer end (the inner chain well configured so as to not allow a normally thrown disc from moving the chains out from the outer chain well).

[0228] In one embodiment, a kit including an adjustable target disc-catching device disclosed herein and a device for dampening the sound of a disc thrown into an adjustable target disc-catching device disclosed herein.

[0229] In one embodiment, the device for dampening the sound is a length of foam with a hollow core, such that the length of foam with a hollow core can be wrapped around the pole above the pole-ring and below the support frame.

[0230] In one embodiment, a method of resolving disputes, the method including stating the dispute and the two possible resolutions, determining the resolution if a player

makes a putt, the player attempting the putt, and judging if the putt was made. If made, the player who made the putt will have the dispute resolved in his favor; if not made, the player who missed the putt will have the dispute resolved in the favor of his opponent.

[0231] In one embodiment, a support frame of an adjustable target disc-catching device, including a plurality of rails dispersed substantially radially, wherein each rail is an length of solid material featuring an outer portion with an outer end, an inner portion with an inner end, wherein the outer end of the rail is attached to an outer chain barrier; the inner end of the rail is attached to an inner chain barrier; and wherein the outer portion of the rail includes an inclining sliding plane. In one embodiment, the inclining sliding plane has an angle and length selected from: i) at least about 30 degrees and at least about 15 cm; ii) at least about 40 degrees and at least about 10 cm; iii) at least about 50 degrees and at least about 8 cm; iv) at least about 60 degrees and at least about 7 cm; or v) at least about 70 degrees and at least about 6 cm. In one embodiment, the inner portion of the rail includes a declining sliding plane. In one embodiment, the inner end of the rail is below the high point of the inclining sliding plane. In one embodiment, the support frame includes a central pole-hub. In one embodiment, the support frame includes a spoke extending radially from the central pole-hub. In one embodiment, the support frame includes a plurality of spokes extending radially from the central pole-hub. In one embodiment, the inner end of the rail is connected to the spoke. In one embodiment, the spoke is connected to an inner ring. In one embodiment, the spoke is connected to an outer ring. In one embodiment, the inner end of the rail connects to the inner ring. In one embodiment, a chain is connected each rail. In one embodiment, the inner chain barrier is the inner ring. In one embodiment, the inner chain barrier is a length of material. In one embodiment, the outer chain barrier is the outer ring. In one embodiment, the outer chain barrier is a length of material.

[0232] In one embodiment, an adjustable target disc-catching device, including a rail, wherein the rail is an unobstructed length of solid material featuring an outer portion including an outer end and an inner portion including an inner end, wherein the outer end of the rail is attached to an outer chain barrier; the inner end of the rail is attached to an inner chain barrier; wherein the outer portion of the rail includes an inclining sliding plane and the inner portion of the rail includes a declining sliding plane. In one embodiment, the rail has a chain attached, such that the chain can reversibly be manually pushed along the rail from the outer end of the rail to the inner end of the rail, while not allowing a normally thrown disc from moving the chain from the outer end of the rail to the inner end of the rail (or vice versa). In one embodiment, an adjustable target disc-catching device features a plurality of rails. In one embodiment, the inclining sliding plane of the rail has an angle and length of: i) at least about 30 degrees and at least about 15 cm; ii) at least about 40 degrees and at least about 10 cm; iii) at least about 50 degrees and at least about 8 cm; iv) at least about 60 degrees and at least about 7 cm; or v) at least about 70 degrees and at least about 6 cm. In one embodiment, the declining sliding plane has a decline in vertical distance from the apex of the inclining sliding plane to the inner chain barrier of at least about 1 mm, at least about 2 mm, at least 3 mm, at least about 4 mm, at least about 5 mm, at least about 6 mm, at least 7 mm, at least about 8 mm, at least about 9 mm, at least about 1 cm, at least 2 cm, at least about 3 cm, at least about 4 cm, at least about 5 cm, at least 6 cm, at least about 7 cm, at least about 8 cm, at least 9 cm, or at

least about 10 cm. In one embodiment, the declining sliding plane has a displacement in horizontal distance from the apex of the inclining sliding plane to the inner chain barrier of at least about 1 mm, at least about 2 mm, at least 3 mm, at least about 4 mm, at least about 5 mm, at least about 6 mm, at least 7 mm, at least about 8 mm, at least about 9 mm, at least about 1 cm, at least 2 cm, at least about 3 cm, at least about 4 cm, at least about 5 cm, at least 6 cm, at least about 7 cm, at least about 8 cm, at least 9 cm, or at least about 10 cm.

[0233] In one embodiment, an adjustable target disc-catching device, including a plurality of evenly distributed rails, wherein each rail is an unobstructed length of solid material featuring an outer portion with an outer end, an inner portion with an inner end, wherein the outer end of the rail is attached to an outer chain barrier; the inner end of the rail is attached to an inner chain barrier; wherein the outer portion of the rail is an inclining sliding plane, and the inner portion of the rail is a declining sliding plane.

[0234] In one embodiment, a support frame for an adjustable target disc-catching device, including a central pole-hub, a plurality of spokes radiating from the central pole-hub, a plurality of evenly distributed rails connecting to the spokes, wherein each rail is a length of solid material featuring an outer portion with an outer end, an inner portion with an inner end, wherein the outer end of the rail is attached to an outer chain barrier; the inner end of the rail is attached to an inner chain barrier; wherein the outer portion of the rail is an inclining sliding plane, and the inner portion of the rail is a declining sliding plane. In one embodiment, the inclining sliding plane of the rail has an angle and length of: i) at least about 30 degrees and at least about 15 cm; ii) at least about 40 degrees and at least about 10 cm; iii) at least about 50 degrees and at least about 8 cm; iv) at least about 60 degrees and at least about 7 cm; or v) at least about 70 degrees and at least about 6 cm.

[0235] In one embodiment, a rail for an adjustable target disc-catching device, including an uninterrupted length of solid material featuring an outer portion with an outer end, an inner portion with an inner end, wherein the outer end of the rail is attached to an outer chain barrier; the inner end of the rail is attached to an inner chain barrier; wherein the outer portion of the rail is an inclining sliding plane, and the inner portion of the rail is a declining sliding plane.

DETAILED DESCRIPTION OF THE FIGURES

[0236] FIG. 1 is a depiction of an adjustable target disc-catching device, including

[0237] a pole (element 1-1);

[0238] a stand affixed to the bottom of the pole (element 1-2);

[0239] an upward facing basket fixed to the pole above the stand (element 1-3);

[0240] a support frame (element 1-4) fixed to the pole above the basket; wherein the support frame includes:

[0241] a central pole-hub (element 1-6),

[0242] a plurality of spokes (element 1-5) extending radially from the central pole-hub,

[0243] an outer ring (element 1-7) connected to the spokes,

[0244] an inner ring (element 1-8) connected to the spokes,

[0245] a stationary chain support (element 1-9) connected to the inner ring;

[0246] two adjustable chain holders (element 1-10A and element 1-10B) dispersed radially on the support frame;

[0247] chains hanging from the adjustable chain holders,

[0248] wherein the adjustable chain holder includes:

[0249] an outer chain well and an inner chain well, wherein the chain hanging from the adjustable chain holder can be reversibly slid from the outer chain well to the inner chain well, or from the inner chain well to the outer chain well.

The adjustable chain holder depicted on the left side of FIG. 1 (element 1-10A) shows a chain in the outer chain well (element 1-11A). The adjustable chain holder depicted on the right side of FIG. 1 (element 1-10B) shows a chain in the inner chain well (element 1-11B). The bottom of these chains are connected to pole-ring (element 1-12B). The bottom of the chain connected to the stationary chain support (element 1-9) is connected to a pole-ring (element 1-12A).

[0250] FIG. 2 depicts non-exhaustive examples of adjustable chain holders. FIG. 2A, FIG. 2B, FIG. 2C, FIG. 2D, FIG. 2E, FIG. 2F, FIG. 2G, FIG. 2H, FIG. 2I, FIG. 2J, FIG. 2K, FIG. 2L, FIG. 2M, FIG. 2N, and FIG. 2P are vertical cross-sectional views. FIG. 2O is a top-down cross-sectional view of an adjustable chain holder where the well-wall extends into the horizontal plane. Elements 2-1 are outer chain wells. Elements 2-2 are inner chain wells. Elements 2-3 are well-walls. Elements 2-4 are inner chain barriers (for example, but not necessarily, the inner ring). Elements 2-5 are outer chain barriers (for example, but not necessarily, the outer ring). Element 2-6 in FIG. 2P is an intermediate chain well (allowing for three positions of chains for three different diameters of target).

[0251] FIG. 3A and FIG. 3B depict two cross sections of adjustable chain holders. Elements 3-1 are the pole. Elements 3-2 are a spoke. Elements 3-3 are the inner chain barrier. Elements 3-4 is a declining sliding plane and is the portion that forms the inner chain well. Elements 3-5 is an inclining sliding plane and is the portion that forms the outer chain well. For the avoidance of doubt, elements 3-4 and elements 3-5 are one length of solid material. The outer portion of elements 3-5 are bounded by the outer chain barrier (elements 3-6); the juncture of element 3-5 and element 3-6 forms the low point of the outer chain well from which the chain hangs. The inner portion of elements 3-4 are bounded by the inner chain barrier (elements 3-3) forming the low point of the inner chain well (from which the chain will hang). In FIG. 3A the chain (element 3-7A) is hanging from the outer chain well (note that the chain is not depicted resting against the outer chain barrier, for clarity concerns of the figure). In FIG. 3B the chain (element 3-7B) is hanging from the inner chain well (note that the chain is not resting against the inner chain barrier, for clarity concerns of the figure).

[0252] FIG. 4 is a depiction of an adjustable target disc-catching device (showing only six adjustable chain holders and six chains for clarity). Element 4-1 is the outer chain well, from which chains are hanging. The outer chain well is bounded by an outer chain barrier that is not the outer ring. Rather, hanging from the outer ring is a vertical leg connected to an inclining sliding plane, and a steeper inclining sliding plane attached to it. The vertical leg serves as the outer chain barrier. The inclining sliding plane is of sufficient length and steepness to be a hit-sliding barrier. Further, the adjustable chain holders are dispersed substantially radially on the support frame, although they are not perfectly parallel to the spokes, but deviate from the spoke by approximately 15 degrees (as further shown by FIG. 5A and FIG. 5B) and attach to the inner ring (one skilled in the art

realizes that more adjustable chain holders results in smaller degree deviations in the attachment to the inner ring). The ajar nature of the adjustable chain holder to the spoke allows space for chains to pass over the adjustable chain holder without interference from the spoke. FIG. 5A, FIG. 5B, and FIG. 6 further show details of this type of connectivity. Element 4-2 is the inner chain well from which chains are hanging. The inner chain barrier is the inner ring.

[0253] FIG. 5A is a perspective depiction of a support frame with spokes (element 5-3), an inner ring, an outer ring, and an adjustable chain holder. Element 5-1 shows the outer chain well. Element 5-2 shows the inner chain well. The inner portion of the adjustable chain holder connects to the inner ring at approximately the midpoint of the inner ring between two adjoining spokes.

[0254] FIG. 5B is a “birdie-eye view” of a support frame with spokes (element 5-3), an inner ring, an outer ring, and adjustable chain holders. The inner portion of the adjustable chain holder connects approximately to the midpoint of the inner ring between two adjoining spokes. Element 5-1 shows the outer chain well. Element 5-2 shows the inner chain well.

[0255] FIG. 6 depicts a cross-section of an example adjustable chain holder, similar to those depicted in FIG. 4, FIG. 5A, and FIG. 5B. Element 6-1 is the outer ring. Element 6-2, which connects to the outer ring, is the outer chain barrier. Element 6-3 is an inclining sliding plane. Element 6-4 is an inclining sliding plane that is at a greater steepness compared to element 6-3. The space above element 6-3 and element 6-4 forms the outer chain well. Element 6-5 is a declining sliding plane. The space above element 6-5 forms the inner chain well. Element 6-6 is the inner chain barrier. Element 6-7 is a spoke, which is connected to a pole (element 6-8). Note that in some embodiments the spoke (element 6-7) may continue to connect to element 6-1, so long as element 6-4 is at a horizontal angle sufficient to allow a chain to be slid from the outer chain well to inner chain well (or vice versa) without interference from the spoke. Without being limited by these examples, in one embodiment element 6-2 is approximately 4.5 cm in length. In one embodiment element 6-3 is approximately 2.5 cm in length with an angle of approximately 20 degrees. In one embodiment element 6-4 is approximately 13 cm in length with an angle of approximately 45 degrees. In one embodiment element 6-5 is approximately 6 cm in length with an angle of approximately 70 degrees. In one embodiment element 6-7 is approximately 11 cm in length.

[0256] FIG. 7 is a depiction of an adjustable target disc-catching device that does not feature an inner ring or outer ring or a central pole-hub. Only two adjustable chain holders are shown for clarity and one skilled in the art knows there could be numerous adjustable chain holders. Elements 7-1 are outer chain barriers. Elements 7-2 are inner chain barriers. Chain (element 7-3) hangs from the inner chain well. Chain (element 7-4) hangs from the outer chain well. For clarity purposes, the chains are not shown hanging in the low point of their respective chain wells. Although only two adjustable chain holders are shown for clarity, one skilled in the art could fashion an adjustable target disc-catching device featuring many additional adjustable chain holders. For example, an adjustable target disc-catching device may have 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, or 20 or more adjustable chain holders.

[0257] FIG. 8A and FIG. 8B have a similar configuration as FIG. 4 but with some differences. Six identical rail systems are dispersed substantially radially above an upward facing basket, and the rail systems connect the inner ring to

the outer ring. One skilled in the art realizes that any suitable rail system can connect the inner ring to the outer ring, for example, any of those depicted in FIG. 2A, FIG. 2B, FIG. 2C, FIG. 2D, FIG. 2E, FIG. 2F, FIG. 2G, FIG. 2H, FIG. 2I, FIG. 2J, FIG. 2K, FIG. 2L, FIG. 2M, FIG. 2N, FIG. 2O, and FIG. 2P. FIG. 8A depicts chains hanging from the outer chain wells. If a disc golfer wanted a narrower target (for example to hone the player's putting skills), the player could slide the chains over the hit-sliding barrier depositing the chains in the inner chain wells. FIG. 8B depicts chains hanging from the inner chain wells. As can be seen, the target area (the chains) is narrower, making a throw-in harder. For example, the width of the chains in FIG. 8A may be about two feet across, but when the chains are slid into the inner chain well (FIG. 8B) the width of the chains may be about a foot across.

[0258] FIG. 9 is a top perspective view of a stand for a disc golf basket showing my new design. The broken lines in the drawings are used to illustrate unclaimed portions of the stand for a disc golf basket and form no part of the claimed design.

[0259] FIG. 10 is a front view of a stand for a disc golf basket showing my new design. The broken lines in the drawings are used to illustrate unclaimed portions of the stand for a disc golf basket and form no part of the claimed design.

[0260] FIG. 11 is a rear view of a stand for a disc golf basket showing my new design. The broken lines in the drawings are used to illustrate unclaimed portions of the stand for a disc golf basket and form no part of the claimed design.

[0261] FIG. 12 is a left view of a stand for a disc golf basket showing my new design. The broken lines in the drawings are used to illustrate unclaimed portions of the stand for a disc golf basket and form no part of the claimed design.

[0262] FIG. 13 is a right view of a stand for a disc golf basket showing my new design. The broken lines in the drawings are used to illustrate unclaimed portions of the stand for a disc golf basket and form no part of the claimed design.

[0263] FIG. 14 is a top view of a stand for a disc golf basket showing my new design.

[0264] The broken lines in the drawings are used to illustrate unclaimed portions of the stand for a disc golf basket and form no part of the claimed design.

[0265] FIG. 15 is a bottom view of a stand for a disc golf basket showing my new design. The broken lines in the drawings are used to illustrate unclaimed portions of the stand for a disc golf basket and form no part of the claimed design.

[0266] FIG. 16 is a top perspective view of a chain-support frame of an adjustable target disc golf basket showing my new design. In one prototype with this design, the vertical height from the low point of the outer chain barrier to the apex of the hit-sliding barrier is about 12 cm. The length of the rail system from the low point of the outer chain barrier to the apex of the hit-sliding barrier is about 17 cm. The vertical height from the inner chain barrier (inner ring) to the apex of the hit-sliding barrier is about 7 cm. The horizontal distance from the inner chain barrier to the outer chain barrier is about 14 cm. The angle of the midpoint of the inclining sliding plane is about 50 degrees. The inclining sliding plane deviates by about 10 degrees from a line extending from the pole. Featuring a chain in each of the twelve outer chain wells, each chain weighing about 150 grams (and containing about 20 links), the disc-entrapment

device described in this paragraph has been used for thousands of putts, and a thrown disc has not yet moved a chain from an outer chain well to an inner chain well.

[0267] FIG. 17 is a front view of a chain-support frame of an adjustable target disc golf basket (a disc-entrapment device). A pole with a diameter of about 4 cm is able to slide under three of the apex of the hit sliding barriers, and above the spokes.

[0268] FIG. 18 is a rear view of a chain-support frame of an adjustable target disc golf basket.

[0269] FIG. 19 is a right view of a chain-support frame of an adjustable target disc golf basket.

[0270] FIG. 20 is a left view of a chain-support frame of an adjustable target disc golf basket.

[0271] FIG. 21 is a top view of a chain-support frame of an adjustable target disc golf basket.

[0272] FIG. 22 is a bottom view of a chain-support frame of an adjustable target disc golf basket.

[0273] FIG. 23 is a top perspective view of a chain-support frame of an adjustable target disc golf basket showing my new design.

[0274] FIG. 24 is a front view of a chain-support frame of an adjustable target disc golf basket.

[0275] FIG. 25 is a rear view of a chain-support frame of an adjustable target disc golf basket.

[0276] FIG. 26 is a right view of a chain-support frame of an adjustable target disc golf basket.

[0277] FIG. 27 is a left view of a chain-support frame of an adjustable target disc golf basket.

[0278] FIG. 28 is a top view of a chain-support frame of an adjustable target disc golf basket.

[0279] FIG. 29 is a bottom view of a chain-support frame of an adjustable target disc golf basket.

[0280] FIG. 30 is a top perspective view of a chain-support frame of an adjustable target disc golf basket showing my new design.

[0281] FIG. 31 is a front view of a chain-support frame of an adjustable target disc golf basket.

[0282] FIG. 32 is a rear view of a chain-support frame of an adjustable target disc golf basket.

[0283] FIG. 33 is a right view of a chain-support frame of an adjustable target disc golf basket.

[0284] FIG. 34 is a left view of a chain-support frame of an adjustable target disc golf basket.

[0285] FIG. 35 is a top view of a chain-support frame of an adjustable target disc golf basket.

[0286] FIG. 36 is a bottom view of a chain-support frame of an adjustable target disc golf basket.

Examples

[0287] Five people were timed moving the chains of a disc-catching device similar to that depicted in FIG. 23 featuring 12 adjustable chain holders. The results are displayed in Table 2. Referencing FIG. 6, the dimensions of the adjustable chain holders are as follows: element 6-2 is approximately 4.5 cm in length; element 6-3 is approximately 2.5 cm in length with an angle of approximately 20 degrees; element 6-4 is approximately 13 cm in length with an angle of approximately 45 degrees; element 6-5 is approximately 6 cm in length with an angle of approximately 70 degrees; and element 6-7 is approximately 11 cm in length. Spoke (element 6-7) continues and connects to outer chain barrier (element 6-1 which is an outer ring). Element 6-5 connected to the inner ring approximately 2.5 cm from a spoke on either side. The line connecting the juncture of element 6-5 and the inner ring and the juncture of elements 6-4 and elements 6-3 intersects the line con-

necting the juncture of elements 6-4 and elements 6-3 and element 6-8 at an angle of about 15 degrees. The line connecting the length of element 6-4 to the line connecting the juncture of elements 6-4 and elements 6-3 and element 6-8 is an angle of about 25 degrees. The outer diameter is approximately 53 cm and the inner diameter is approximately 26 cm. That is, the target width when chains are in the outer chain well is approximately 53 cm, but when the chains are slid into the inner chain well, the target width is approximately 26 cm (reducing the size of the target, causing the player to aim better in hopes landing a shot in the basket). The disc-entrapment device described in this paragraph has been used for thousands of putts, and a thrown disc has not yet moved a chain from an outer chain well to an inner chain well.

[0288] Four of the participants, Person 2, Person 3, Person 4, and Person 5, had never previously touched the disc-catching device. Person 3, Person 4, and Person 5 had never previously played disc golf. All participants used two hands except for Person 3 who used only one hand. The data demonstrates that it is easy, fast, and simple, even for a novice, to use the adjustable target disc-catching device disclosed herein.

TABLE 2

Data.		
Participant	Time to move all chains from outer chain wells to inner chain wells	Notes
Person 1	7.2 seconds	Inventor
Person 2	9.1 seconds	Wife of inventor, timed during her first time moving chains (novice chain mover)
Person 3	14.6 seconds	Timed during first time moving chains (novice chain mover) - used only one hand, holding a cappuccino in the other hand
Person 4	7.2 seconds	Cousin of Person 2 timed during her first time moving chains (novice chain mover)
Person 5	7.6 seconds	Fiancé of Person 4 timed during his first time moving chains (novice chain mover)

[0289] Furthermore, although the foregoing has been described in some detail by way of illustrations and examples for purposes of clarity and understanding, it will be understood by those of skill in the art that numerous and various modifications can be made without departing from the Platonic Form of Ideal Adjustable Target Disc-Catching Device of the present disclosure. Therefore, it should be clearly understood that the forms disclosed herein are illustrative only and are not intended to limit the scope of the present disclosure, but rather to also cover all modifications and alternatives coming with the true scope, spirit, and Platonic Form of the disclosure provided herein.

1. A support frame for an adjustable target disc-catching device, comprising:

- a central pole-hub;
- a plurality of spokes radiating from the central pole-hub;
- a plurality of evenly distributed rails connecting to the spokes;
- wherein each rail is a length of solid material comprising:
 - an outer portion with an outer end;
 - an inner portion with an inner end;
 - wherein the outer end of the rail is attached to an outer chain barrier;
 - the inner end of the rail is attached to an inner chain barrier;
 - wherein the outer portion of the rail comprises an inclining sliding plane; and
 - the inner portion of the rail comprises a declining sliding plane.

2. The support frame of claim 1, wherein the inclining sliding plane of the rail has an angle and length selected from:

- i) at least about 30 degrees and at least about 15 cm;
- ii) ii) at least about 40 degrees and at least about 10 cm;
- iii) iii) at least about 50 degrees and at least about 8 cm;
- iv) iv) at least about 60 degrees and at least about 7 cm;
- or
- v) v) at least about 70 degrees and at least about 6 cm.

3. The support frame of claim 1, wherein a chain hangs from the rail.

* * * * *